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CHAPTER 21

USING FORMATTING, EDITING FEATURES OF WORD PROCESSORS FOR PRODUCING DOCUMENTS

SUGGESTED TIME: 7 PERIODS

IN THIS CHAPTER

This chapter develops advanced mastery of word processors for producing well-formatted documents, then builds complete competence in spreadsheets: formatting data, cell referencing, formulae, functions, charts, linked worksheets and what-if analysis. It prepares learners for the practical and theory papers of the GCE Advanced Level in ICT.

- By the end of this chapter I can produce a professional document using formatting and editing features of a word processor (styles, tables, mail merge, headers/footers).
- By the end of this chapter I can enter, format, sort, filter and manage data in a spreadsheet.
- By the end of this chapter I can distinguish relative, absolute and mixed cell references and use them correctly in formulae.
- By the end of this chapter I can write basic formulae and use advanced functions (IF, VLOOKUP, SUMIF, COUNTIF, statistical functions).
- By the end of this chapter I can create and customise charts that correctly represent worksheet data.
- By the end of this chapter I can link data across multiple worksheets and workbooks.

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COURSE complete, illustrated, step by step

METHODS and worked examples

EXERCISES exam style + solutions

SUMMARY SHEET the essentials to remember

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CONTENTS

1. Producing Documents with Word Processing Features
2. Formatting and Managing Spreadsheet Data
3. Cell Referencing and Basic Formulae
4. Advanced Functions, Charts and Linked Worksheets
5. What-If Analysis and Integration Activities
6. Integration activity
7. Methods and worked examples
8. Exercises
9. Solutions to the exercises
10. Summary sheet

CHAPTER OBJECTIVES

- By the end of this chapter I can produce a professional document using formatting and editing features of a word processor (styles, tables, mail merge, headers/footers).
- By the end of this chapter I can enter, format, sort, filter and manage data in a spreadsheet.
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- By the end of this chapter I can create and customise charts that correctly represent worksheet data.
- By the end of this chapter I can link data across multiple worksheets and workbooks.
- By the end of this chapter I can perform what-if analysis using Goal Seek, scenarios and data tables.
- By the end of this chapter I can integrate word processing and spreadsheet outputs into a single professional report.

PREREQUISITES

- Basic computer operations: files, folders, saving and opening documents (Lower Sixth).
- Introduction to word processing: typing, selecting, copying and pasting text.
- Introduction to spreadsheets: cells, rows, columns, worksheets and simple data entry.
- Basic mathematical operations and percentages.
- Keyboard and mouse handling skills acquired in Lower Sixth ICT practical sessions.

1 PRODUCING DOCUMENTS WITH WORD PROCESSING FEATURES

At the end of every term, the bursar of a secondary school in Bamenda spends three days retyping fee records, recalculating totals by hand, and redrawing the same tables in Word. One wrong figure in a receipt book can trigger a dispute with a parent at the next PTA meeting. Meanwhile, a cybercafé operator in Douala produces the same reports in thirty minutes using Word formatting and Excel formulas that update themselves. What does he know that the bursar does not? He knows that a **word processor** is not a typewriter: it is a document-production machine in which formatting is structured, repetitive work is automated, and one master document can generate hundreds of personalised copies. This section turns you into that efficient operator. We begin with the tool every GCE candidate must master: the word processor (Microsoft Word being the reference software in the syllabus).

1.1 The Word Processor Interface: Knowing Your Workshop

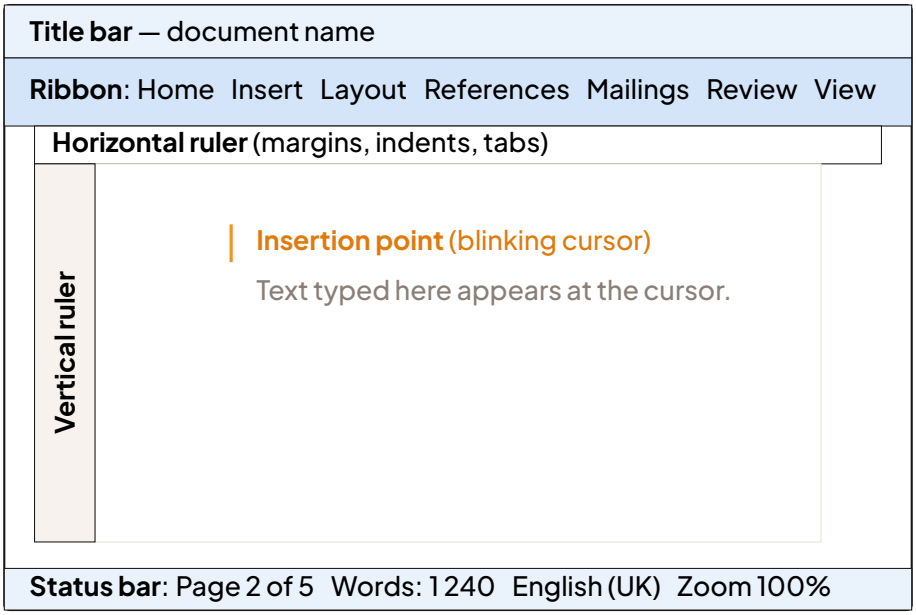
A carpenter who cannot find his tools wastes the day. The same is true of a Word user who cannot read the screen. When Word opens, the window presents several zones, each with a precise role.

DEFINITION **Key zones of the Word window**

The **ribbon** is the strip of tabs (Home, Insert, Layout, References, Mailings, Review, View) at the top of the window; each tab groups related commands into groups. The **ruler** (horizontal and vertical) shows margins, indents and tab stops. The **insertion point** is the blinking vertical bar showing where the next typed character will appear. The **status bar**, at the bottom, displays the current page number, word count, language and zoom level.

Word also offers several **views**, chosen from the View tab or the status bar:

- **Print Layout view**: shows the document exactly as it will be printed (margins, headers, page breaks visible). This is the default working view.
- **Draft view**: hides headers, footers and page boundaries so you can focus on the text itself; useful for fast typing and editing.
- **Read Mode, Web Layout and Outline views**: for reading on screen, previewing web pages, and reorganising long documents by headings respectively.



Mock-up of a Word window showing the ribbon, rulers, insertion point and status bar.

WARNING **Do not confuse the ruler with margins**

Dragging the small markers on the ruler changes indentation for the selected paragraphs, not the page margins. Margins are set in Layout → Margins. Mixing the two up is a classic source of crooked documents.

1.2 Character Formatting

Character formatting applies to individual letters, words or any selected text. All of it lives in the **Font** group of the Home tab.

- **Font (typeface)**: the design of the characters, e.g. Times New Roman (formal documents), Calibri, Arial. Choose one readable font and stick to it.
- **Size**: measured in points (pt). Body text is usually 11 or 12 pt; headings are larger.
- **Bold, Italic, Underline** (Ctrl+B, Ctrl+I, Ctrl+U): used for emphasis. Underline is largely reserved for hyperlinks in modern documents.
- **Superscript and subscript**: characters raised above or lowered below the baseline, e.g. x^2 (superscript) or H_2O (subscript). Essential in science reports.
- **Text effects**: strikethrough, small caps, highlight colour, font colour, text outline and shadow.

EXEMPLE **Formatting a chemistry note**

To type "The formula of water is H_2O and $E = mc^2$", select the "2" in H2O and click the subscript button (or Ctrl+=); select the "2" in mc^2 and click superscript (Ctrl+Shift+=). The formatting applies only to the selected characters.

1.3 Paragraph Formatting

Paragraph formatting applies to whole paragraphs (a paragraph ends wherever the Enter key was pressed). It is found in the **Paragraph** group of the Home tab and in the Paragraph dialog box.

- **Alignment**: left (default), centred, right, or justified (both edges straight — used in newspapers and official letters).
- **Line spacing**: the vertical space between lines (1.0, 1.5, 2.0…). GCE reports are often required in 1.5 spacing for readability.
- **Indentation**: the distance of a paragraph from the margin. The first-line indent starts the first line further right; a hanging indent does the opposite (used in bibliographies).
- **Spacing before/after**: space inserted between paragraphs without pressing Enter.
- **Bullets and numbering**: automatic lists. Numbering renumbers itself when items are inserted or deleted — a huge advantage over typing "1.", "2." by hand.
- **Borders and shading**: lines around, and background colour behind, a paragraph — useful for highlighting a warning or a title.

WARNING **The Enter-key disease**

Beginners press Enter several times to create space between paragraphs or to jump to a new page. This is wrong: the moment you add or delete text above, the whole layout collapses and pages shift unpredictably. Use paragraph spacing for gaps and Layout → Breaks → Page Break (Ctrl+Enter) to force a new page. A page break is a solid instruction; empty paragraphs are a house of cards.

1.4 Page Setup: Margins, Orientation, Size, Columns and Breaks

Before printing, the page itself must be configured from the **Layout** tab:

- **Margins**: the white space around the text (Normal = 2.54 cm on all sides). Official letters often use wider left margins for binding.
- **Orientation**: Portrait (taller than wide, default) or Landscape (wider than tall, good for large tables and certificates).
- **Paper size**: A4 is the standard in Cameroon; Letter is the American size.

- **Columns**: text can flow in two or three newspaper-style columns (Layout → Columns).
- **Breaks**: a page break starts a new page; a section break starts a new section, which can have its own orientation, margins, columns and headers. For example, one landscape page holding a wide table inside a portrait report requires a section break before and after it.

1.5 Headers, Footers, Page Numbers, Footnotes and Endnotes

The **header** is the repeated zone at the top of every page; the **footer** is its counterpart at the bottom. Double-click the top or bottom of a page (or Insert → Header/Footer) to edit them. Typical content: school name, document title, date. **Page numbers** (Insert → Page Number) update automatically — you never type them by hand.

A **footnote** is a note printed at the bottom of the page where its reference mark appears; an **endnote** is collected at the end of the document. Both are inserted via References → Insert Footnote/Endnote and are numbered automatically. They are used to cite sources or add side remarks without interrupting the text.

1.6 Styles, Themes and the Automatic Table of Contents

Here lies the single most powerful idea of this section: never format headings by hand.

DEFINITION **Style**

A **style** is a named set of formatting characteristics (font, size, colour, spacing, alignment) that is applied consistently to text. Examples: Heading 1, Heading 2, Title, Normal. A **theme** is a coordinated set of colours and fonts applied to the whole document at once.

Why are styles so important? First, **consistency**: every chapter heading looks identical. Second, **instant change**: modify the style once and every heading in a 60-page report updates at once. Third — and this is decisive — Word uses heading styles to build an automatic table of contents.

METHOD **Creating an automatic table of contents**

1. Apply Heading 1 to each chapter title and Heading 2 to each sub-title (Home tab, Styles gallery).
2. Click where the table of contents must appear (usually after the title page).
3. Go to References → Table of Contents and choose an automatic style.
4. After editing the document, right-click the table and choose Update Field → Update entire table so titles and page numbers are refreshed.

1.7 Tables in Word

A **table** organises information in rows and columns; the intersection of a row and a column is a cell. Insert one via Insert → Table (choose the grid size, or Insert Table for exact dimensions). Key operations:

- **Merging cells**: select several cells, right-click → Merge Cells — used for a title spanning several columns.
- **Splitting cells**: divides one cell into several rows or columns.
- **Sorting**: Table Tools → Layout → Sort arranges rows alphabetically or numerically by any column (e.g. class list by marks, descending).
- **Simple formulas**: Table Tools → Layout → Formula accepts expressions such as `=SUM(ABOVE)` or `=AVERAGE(LEFT)` to total a column of figures — handy for a fee table, though a spreadsheet is the proper tool for heavy calculations (next sections).

WORKED EXAMPLE 1 **Producing a fee summary table**

The bursar must present a table of fees collected: Form Five 4 500 000 FCFA, Lower Sixth 3 800 000 FCFA, Upper Sixth 4 100 000 FCFA, with a total row.

SOLUTION

Solution.

1. Insert → Table, 2 columns × 5 rows.
2. Row 1: type the headers Class and Amount (FCFA); make them bold and shaded (Table Design tab).
3. Rows 2–4: enter the three classes and amounts.
4. Row 5, cell 1: type Total. Merge nothing needed here.
5. Click in the last cell, Table Tools → Layout → Formula, accept `=SUM(ABOVE)`: Word displays 12 400 000.
6. To rank classes, select rows 2–4 and use Sort on column 2, descending.

Note: if a figure changes, the Word total does not update automatically — select it and press F9 to refresh the field.

1.8 Editing Features: Find and Replace, Proofing, Track Changes

Professional documents are edited, not merely typed.

- **Find** (Ctrl+F) locates a word or phrase. **Replace** (Ctrl+H) substitutes one text for another throughout the document — e.g. replacing "2023" by "2024" in a reused circular, or fixing a misspelt name everywhere at once.

- **Spelling and grammar check** (Review → Spelling & Grammar, or F7): red wavy underlines signal spelling errors, blue/green underlines signal grammar issues. Right-click an underlined word for suggestions.
- **Track Changes** (Review tab): records every insertion and deletion so another person can accept or reject them. **Comments** attach margin notes to selected text without altering it. A supervisor correcting a student's project report uses exactly these two tools.

1.9 Mail Merge: One Letter, Five Hundred Parents

Back to our bursar. He must send a fee-reminder letter to 500 parents. Each letter is identical except the parent's name, the child's name, the class and the balance due. Retyping 500 letters is three days of work and a guarantee of errors. **Mail merge** does it in minutes.

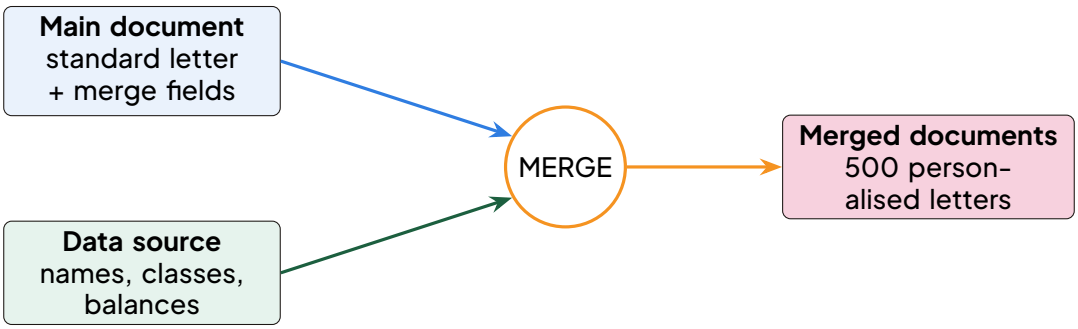
DEFINITION **Mail merge**

Mail merge is a word-processing feature that combines a standard document with a data source to produce many personalised copies automatically.

KEY POINTS **The three components of mail merge — classic GCE question**

1. The **main document**: the standard letter containing the fixed text and the merge fields (placeholders such as «ParentName »).
2. The **data source**: the table of variable information (names, classes, balances) — often an Excel worksheet, an Access table or a Word table.
3. The **merged document**: the final personalised output (500 individual letters), which can be printed, saved or sent by e-mail.

Learn to list and define these three components: it is one of the most frequent theory questions on this topic.



The mail merge process: the main document and the data source are combined to produce personalised merged documents.

METHOD **Producing personalised fee-reminder letters**

1. Type the standard letter in Word (the main document).
2. Prepare the data source: an Excel sheet with columns ParentName, ChildName, Class, Balance.
3. In Word: Mailings → Start Mail Merge → Letters.
4. Select Recipients → Use an Existing List, and choose the Excel file.
5. Place the cursor where each variable belongs and click Insert Merge Field (e.g. «ParentName», «Balance»).
6. Click Preview Results to check a few records.
7. Finish & Merge → Print Documents (or Edit Individual Documents to save all 500 letters in one file).

DID YOU KNOW? **Mail merge in Cameroonian life**

Banks, insurance companies and mobile money operators use mail-merge logic daily: the SMS you receive saying "Dear Ngong Paul, your balance is…" is generated by merging a standard message with a database of millions of customers. The same principle powers the personalised GCE registration slips printed for each candidate.

1.10 Printing and Exporting to PDF

Before printing, always use **File → Print** and inspect the print preview: check the page range, number of copies, printer, and whether to print one or both sides (duplex). To share a document that must not be modified and must look identical on every device, **export it to PDF** (File → Save

As / Export → PDF). PDF preserves fonts, layout and images exactly — it is the format expected for official circulars and for submitting typed projects.

EXERCISE 1 **Fee reminder project**

The principal of GBHS Bamenda asks you to send fee reminders to the parents of 40 Upper Sixth students.

1. Name the three components of the mail merge you will perform, and state the role of each.
2. Give two reasons why an automatic table of contents is preferable to a typed one in the end-of-year report.
3. Explain why inserting a page break is better than pressing Enter repeatedly at the end of a chapter.

SOLUTION **Exercise 1**

1. Main document: the standard reminder letter with merge fields; data source: the table (e.g. Excel sheet) holding the 40 parents' names, students' names, classes and balances; merged document: the 40 personalised letters produced by combining the two.
2. (a) It updates automatically (Update Field) when titles or pages change; (b) it is generated from heading styles, so page numbers and titles are always exact, with no typing errors.
3. A page break is a fixed instruction: the next chapter always starts on a fresh page whatever edits occur above. Repeated Enters create empty paragraphs that shift whenever text is added or deleted, destroying the layout.

KEY POINTS **Section summary**

- The Word window: ribbon, rulers, insertion point, status bar; Print Layout vs Draft view.
- Character formatting (font, size, bold/italic/underline, super/subscript) vs paragraph formatting (alignment, spacing, indentation, bullets, borders).
- Page setup: margins, orientation, paper size, columns, page and section breaks.
- Headers/footers, automatic page numbers, footnotes and endnotes.
- Styles guarantee consistency and feed the automatic table of contents.

- Word tables: merge/split cells, sort, simple formulas such as `=SUM(ABOVE)`.
- Editing tools: Find/Replace, spelling and grammar, track changes and comments.
- Mail merge = main document + data source → merged documents; finish with printing or PDF export.

2 FORMATTING AND MANAGING SPREADSHEET DATA

In the previous section we used a word processor to present information: letters, reports and tables that are essentially fixed once printed. But a school report card, a fee-collection record or a stock list at a Douala market is never fixed: values change, new rows appear, and we constantly need to rank, filter and check the data. This is exactly why the **spreadsheet** exists. A spreadsheet such as Microsoft Excel, LibreOffice Calc or Google Sheets stores data in a grid where every value can be formatted, sorted, filtered and checked automatically. In this section we learn to enter data correctly, format it professionally and manage it efficiently — the foundations on which formulae, charts and what-if analysis (next sections) will rest.

2.1 Spreadsheet Structure Revisited

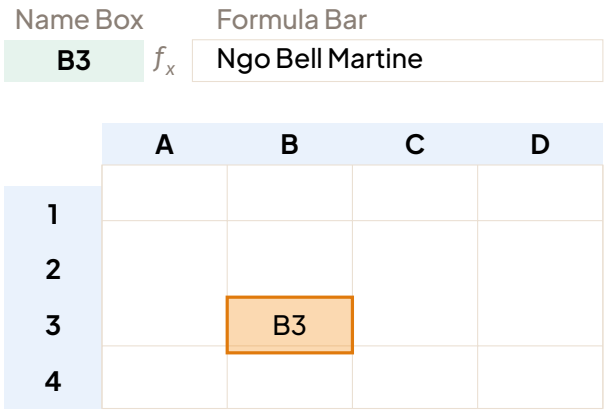
DEFINITION **Workbook and worksheet**

A **workbook** is the spreadsheet file itself (for example `Term2_Results.xlsx`). A **worksheet** is a single grid of rows and columns within a workbook file. A workbook may contain several worksheets, shown as tabs at the bottom of the window (e.g. “Form 5”, “Lower Sixth”, “Upper Sixth”).

The intersection of a row and a column is a **cell**. Columns are labelled with letters (A, B, C, …, Z, AA, AB, …) and rows with numbers (1, 2, 3, …). Each cell therefore has a unique **cell address** (or reference): the column letter followed by the row number, for example `B3`. A rectangular group of cells is a **range**, written with a colon: `A1:D10` means all cells from `A1` to `D10`.

Two screen elements are essential:

- the **Name Box** (left of the formula bar) shows the address of the active cell;
- the **Formula Bar** shows the true content of the active cell — which may differ from what the cell displays after formatting.



A worksheet grid: the active cell `B3` is the intersection of column B and row 3. The Name Box confirms the address; the Formula Bar shows the real content.

WARNING **Displayed value vs real content**

Formatting changes only the display. If a cell contains 12500.5 and is formatted as currency with no decimals, it shows `12 501 FCFA` but the stored value is still 12500.5. Always check the Formula Bar when a result surprises you.

2.2 Data Types and Correct Entry

A spreadsheet does not treat everything alike: it recognises several **data types**, and the type determines what you can do with the value (calculate, sort chronologically, etc.).

Type	Examples	Correct entry	Default alignment
Text	Names, towns, codes	Typed directly	Left
Number	15, 12500.5, -3	Digits, one decimal point or comma per regional settings	Right
Date	12/01/2026	`12/01/2026` or `12-Jan-2026`	Right
Currency	12 500 FCFA	Type the number, then apply the currency format	Right
Percentage	85 %	Type `85%` or 0.85 with % format	Right

WARNING **Numbers typed as text — the classic trap**

If you type an apostrophe before a number (`'12500`), or add a letter or a space inside it (`12500F`), the spreadsheet stores it as text: it stays left-aligned and Excel shows a small

green triangle in the corner of the cell. Such "numbers" are ignored by calculations — a SUM over them gives a wrong total. The cure: re-enter the value as a pure number, or use the warning menu "Convert to Number".

KEY POINTS **Entry rules**

- One piece of information per cell (do not type "Etoudi Paul 14/20" in one cell).
- Dates are stored internally as numbers, so they can be sorted and subtracted.
- Use the format for FCFA and %, never type the unit inside the cell.

2.3 Cell Formatting

Formatting makes a sheet readable and professional. Select the cells, then use the **Home** tab or right-click → **Format Cells**.

Number formats. The Number category controls how values appear:

- **Currency**: `12 500 FCFA` — choose the symbol and the number of decimal places;
- **Percentage**: the stored value 0.85 displays as `85 %`;
- **Date**: the same stored date can display as `12/01/2026`, `12-Jan-26` or `Monday 12 January 2026`.

Alignment. Horizontal (left, centre, right) and vertical alignment, plus Wrap Text (long text stays inside its cell on several lines) and Merge & Centre (one title across several columns — use sparingly, merged cells complicate sorting).

Borders and fill colours. Borders delimit the table (outside thick border, thin inside lines); fill colours distinguish headers or categories. Choose sober, contrasting colours: a report card is not a carnival poster.

Cell styles. A **cell style** is a named bundle of formats (font, fill, border, number format) applied in one click, e.g. the built-in "Heading 1" or "Total" styles. Styles guarantee consistency across a whole workbook.

EXEMPLE **Formatting a marks table**

For the Upper Sixth class council, format the column of averages as Number, 2 decimals; the column of fees paid as Currency, FCFA, 0 decimals; give the header row a dark blue fill with bold white centred text; and put a thick border around the whole table. The sheet is immediately readable by the principal.

2.4 Adjusting Rows and Columns

Data rarely fits the default grid. Right-click a column letter or row number (or use **Home → Format**) to:

- **Width / Height**: set an exact size, or double-click the boundary between two headers for **AutoFit** (the column takes the width of its longest content);
- **Insert / Delete**: inserting a column shifts the others right; inserting a row shifts rows down. Existing formulae adjust automatically;
- **Hide / Unhide**: hiding a column (e.g. raw test marks) keeps its data and formulae active but removes it from view and from printing. Select the neighbouring columns and choose Unhide to bring it back.

WARNING **##### is not an error**

When a numeric column is too narrow, the cell shows `####`. The value is intact: simply widen the column. (Text, by contrast, is merely truncated visually.)

2.5 AutoFill and Custom Series

Observe the small square at the bottom-right corner of the active cell: the **fill handle**. Dragging it copies or extends the content intelligently.

PROPERTY **AutoFill series recognition**

AutoFill automatically recognises and extends built-in series such as Monday, Tuesday, … and January, February, …, as well as numeric patterns: dragging two selected cells containing 2 and 4 produces 6, 8, 10, … ; dragging a date produces the following days. A single number dragged alone is simply copied (hold Ctrl to increment instead).

EXEMPLE **Preparing a term planner**

Type `Week 1` in A1 and drag the fill handle down: the sheet writes `Week 2`, `Week 3`, … Type `January` in B1 and drag right: `February`, `March`, … appear. You can also define custom lists (File → Options → Advanced → Edit Custom Lists), e.g. the sequence of your school's classes: Form 5, Lower Sixth, Upper Sixth.

2.6 Sorting Data

Sorting rearranges the rows of a table according to the values of one or several columns, without altering the data itself.

METHOD **Sorting on multiple keys**

To rank students by class, then by merit inside each class:

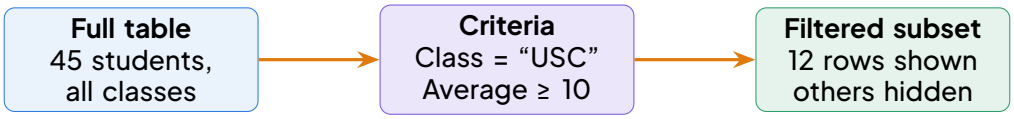
1. Click any single cell inside the table (do not select only one column — that would scramble the rows!).
2. Open Data → Sort.
3. Tick My data has headers so the header row stays on top.
4. First level: Sort by Class, order A to Z.
5. Click Add Level: Then by Average, order Largest to Smallest.
6. Optionally add a third level (e.g. Name, A to Z) to break ties. Click OK.

WARNING **The selection trap**

If you select only the "Average" column before sorting, the spreadsheet may sort that column alone, detaching the averages from the students' names — a disaster for a report card. Always click one cell inside the table and let the software detect the whole range, or select the entire table including all columns.

2.7 Filtering Data with AutoFilter

Sorting reorders; **filtering** hides the rows you do not want to see. With **Data → Filter** (AutoFilter), a drop-down arrow appears on each header. Untick values to hide them, or open Number Filters / Text Filters for **custom criteria**: Greater Than, Between, Contains, …



The filtering process: criteria applied to the full table reveal only the matching subset; hidden rows are not deleted.

EXEMPLE **Custom filter**

To see only students with an average between 10 and 14: open the arrow on the Average header → Number Filters → Between… → enter 10 and 14 → OK. Combine it with Class = USC on another column: filters on several columns cumulate (logical AND). Use Data →

Clear to remove all filters.

WARNING **Filtered data is hidden, not deleted**

Rows hidden by a filter still exist. But beware: copying a filtered range copies only the visible cells, and printing a filtered sheet prints only the visible rows. Check the row numbers (blue, non-consecutive) to know a filter is active.

2.8 Data Validation

The best error is the one that can never be typed. **Data validation** (**Data → Data Validation**) restricts what a cell accepts:

- **List**: the cell offers a drop-down, e.g. `USC;LSC;Form 5` — impossible to misspell a class name;
- **Whole number / Decimal between**: e.g. a mark must lie between 0 and 20;
- **Date between**: e.g. only dates of the current school year.

You can add an Input Message (guidance shown when the cell is selected) and an Error Alert (message shown when the rule is broken).

EXEMPLE **Protecting a mark-entry sheet**

Select the mark column → Data Validation → Allow: Decimal, between 0 and 20 → Error Alert: "A mark must be between 0 and 20." A teacher who types 75 by mistake is stopped immediately instead of corrupting the term averages.

2.9 Conditional Formatting

DEFINITION **Conditional formatting**

Conditional formatting is formatting (colour, font, icon) applied automatically to a cell whenever its value satisfies a stated condition, and removed automatically when the condition ceases to hold.

Via **Home → Conditional Formatting** you can use:

- **Highlight Cells Rules**: Greater Than, Less Than, Between, Equal To, Text that Contains;
- **Top/Bottom Rules**: Top 10 items, Below Average;
- **Data Bars, Colour Scales, Icon Sets**: visual heat-maps of a whole column.

EXEMPLE **Marks below 10 in red**

Select the averages column → Conditional Formatting → Highlight Cells Rules → Less Than → type 10 → choose "Light Red Fill with Dark Red Text". Every weak average turns red instantly; if a mark is corrected and the average rises above 10, the red disappears by itself.

WARNING **Conditional formatting is not a calculation**

Conditional formatting only colours cells; it does not change values and cannot replace a formula. Also, piling up many contradictory rules on the same range makes the sheet unreadable — keep two or three meaningful rules.

2.10 Managing Large Sheets: Freezing, Splitting, Printing

Freeze panes. In a sheet of 300 students, the header row disappears when you scroll down. **View → Freeze Panes → Freeze Top Row** (or Freeze First Column) keeps headers visible while the rest scrolls. To freeze both, select cell `B2` and choose Freeze Panes: everything above and left of `B2` stays fixed.

Split. View → Split divides the window into independent panes with separate scroll bars — useful to compare the top and the bottom of a long list simultaneously.

Page setup and print area. Before printing (**Page Layout** tab):

- choose **Orientation** (landscape for wide tables) and **paper size** (A4);
- **Scale** the sheet to fit one page wide;
- define the **Print Area** (select the range → Print Area → Set Print Area) so that only the useful table prints, not your scratch calculations;
- use Print Titles to repeat the header row on every page.

DID YOU KNOW? **Spreadsheets in Cameroonian schools**

Many colleges and high schools in Cameroon now prepare term report cards with Excel templates: marks are typed once, averages and ranks are computed automatically, and the honour roll is produced by filtering and sorting exactly as in this section. Mastering these skills is directly useful for any secretarial, accounting or teaching career.

WORKED EXAMPLE 2 **Preparing the honour roll of Upper Sixth**

The sheet `Results` contains, in columns A to D: Name, Class, Average, Fees Paid (FCFA)

for 45 students. (1) Format the sheet professionally. (2) Produce the list of USC students ranked by decreasing average, showing only those with an average of at least 10, with failing averages of the whole sheet highlighted in red. (3) Prepare printing of the honour roll alone.

SOLUTION

Solution.

1. Formatting. Select A1:D1 → bold, white text, dark blue fill, centred. Select C2:C46 → Format Cells → Number, 2 decimals. Select D2:D46 → Currency, symbol FCFA, 0 decimals. Select A1:D46 → All Borders, then a thick outside border. AutoFit all column widths (double-click the header boundaries).
2. Red highlight. Select C2:C46 → Conditional Formatting → Less Than 10 → red fill.
3. Filtering. Click inside the table → Data → Filter. On the Class arrow, tick only USC. On the Average arrow → Number Filters → Greater Than Or Equal To 10.
4. Sorting. Data → Sort → Sort by Average, order Largest to Smallest → OK. (Filtering and sorting combine safely.)
5. Printing. Select the visible table → Page Layout → Print Area → Set Print Area; orientation Portrait; Scale: Fit to 1 page wide; Print Titles: repeat row 1. Print Preview to verify, then print.

The result is a clean, ranked honour roll of USC students with averages ≥ 10, ready for the notice board.

EXERCISE 2 **School canteen stock**

The canteen of a college in Bafoussam keeps a sheet with columns: Item, Category, Quantity, Unit Price (FCFA), Expiry Date.

1. Explain how to guarantee that Category contains only “Drink”, “Food” or “Soap”.
2. Describe how to display only items whose quantity is below 20, sorted by expiry date, earliest first.
3. The Quantity column shows green triangles and the total is wrong. Diagnose and fix.
4. State how to keep the header row visible while scrolling through 500 items.

SOLUTION **Exercise 1**

1. Select the Category column → Data Validation → Allow: List → Source:

`Drink,Food,Soap`. A drop-down appears and any other entry is rejected with an error alert.

2. Data → Filter; on Quantity: Number Filters → Less Than 20; then Data → Sort by Expiry Date, order Oldest to Newest (dates sort chronologically because they are true date values).
3. The quantities were entered as text (left-aligned, green triangles), so SUM ignores them. Fix: select the column, use the warning icon → Convert to Number (or retype the values), then recompute the total.
4. Click in row 2 (or any cell below the headers) → View → Freeze Panes → Freeze Top Row.

KEY POINTS **Key points of the section**

- A workbook contains worksheets; a cell is identified by its address (column letter + row number); the Formula Bar shows the true content.
- Enter data in the correct type; apply currency and percentage through formats, never by typing symbols.
- AutoFill extends days, months and numeric series via the fill handle.
- Sort on several keys with Data → Sort → Add Level; filter with AutoFilter and custom criteria — hidden rows are not deleted.
- Data validation prevents wrong entries; conditional formatting colours cells automatically according to rules.
- Freeze panes for large sheets; set the print area, orientation and print titles before printing.

With clean, well-formatted and well-managed data, we are now ready to make the spreadsheet calculate: the next section introduces cell referencing and basic formulae.

3 CELL REFERENCING AND BASIC FORMULAE

In the previous section we learned how to enter, format and manage data in a worksheet: adjusting column widths, applying number formats, sorting and filtering. But a well-formatted table that only displays numbers is no better than a sheet of paper. The real power of a spreadsheet such as Microsoft Excel or LibreOffice Calc is that it computes. Once you tell the spreadsheet how to compute, it does the work for you — and redoes it instantly whenever the data change. The key that unlocks this power is the **formula**, and the ingredient that makes formu-

lae so flexible is the **cell reference**. This section builds both ideas rigorously, from the structure of a formula to the subtle but examinable difference between relative, absolute and mixed references.

3.1 The Structure of a Formula

Imagine a shopkeeper in Mokolo market who sells 15 bags of groundnuts at 2500 FCFA each. On paper she writes 15×2500. In a spreadsheet, the quantity 15 is typed in cell `B2` and the price 2500 in cell `C2`. Instead of typing the result 37500 by hand (which would become wrong the moment a value changes), she types in `D2`:

```
=B2*C2
```

The spreadsheet reads the values stored in `B2` and `C2`, multiplies them, and displays the result. This is a formula.

DEFINITION **Formula**

A **formula** is an expression entered in a cell that always begins with the equal sign `=` and instructs the spreadsheet to perform a calculation. A formula is built from **operands** (constants such as 5, or cell references such as `B2`) combined by **operators**.

DEFINITION **Cell reference**

A **cell reference** is the address of a cell (column letter followed by row number, e.g. `B2`) used inside a formula so that the formula works with the content of that cell rather than with a fixed number. A range of cells is written with a colon, e.g. `B2:B10` means all the cells from `B2` down to `B10`.

The arithmetic operators available in spreadsheet formulae are:

Operator	Meaning	Example (result)
+	Addition	`=2+3` gives 5
-	Subtraction	`=10-4` gives 6
*	Multiplication	`=6*7` gives 42
/	Division	`=20/4` gives 5
^	Exponentiation (power)	`=2^3` gives 8
%	Percentage (divides by 100)	`=19.25%` gives 0.1925

WARNING **The * is compulsory**

In mathematics we write $3x$ to mean $3 \times x$. A spreadsheet does not accept this: `=3A1` is in-

valid. You must always write the multiplication sign explicitly: `=3*A1`. Similarly, the power symbol is the caret `{}`, not a superscript: x^2 is written `=A1{}2`.

3.2 Order of Operations and Parentheses

A formula such as `=2+3*4` could give 20 (if we added first) or 14 (if we multiplied first). Spreadsheets follow the same convention as mathematics, often remembered as BODMAS/PEMDAS:

1. **Parentheses** first;
2. **Exponentiation** (`{}`);
3. **Multiplication and division** (`*`, `/`), from left to right;
4. **Addition and subtraction** (`+`, `-`), from left to right.

Thus `=2+3*4` gives 14, whereas `=(2+3)*4` gives 20. Parentheses override the default priority and make your intention unambiguous.

EXEMPLE **Computing a weighted average**

A student scores 12/20 in ICT (coefficient 4) and 15/20 in Mathematics (coefficient 2), stored in `B2` and `C2`. The weighted total is:

`=B2*4 + C2*2` gives $12 \times 4 + 15 \times 2 = 78$.

To compute the average we must divide by the sum of coefficients, and parentheses are essential:

$$\texttt{=(B2*4 + C2*2)/(4+2)} = \frac{78}{6} = 13.$$

Without parentheses, `=B2*4+C2*2/4+2` would give $48 + 7.5 + 2 = 57.5$ — completely wrong.

PROPERTY **Automatic recalculation**

A formula always recalculates automatically whenever any cell it refers to is modified. If the price in `C2` changes from 2500 to 3000, the formula `=B2*C2` immediately displays the new result 45000. This property is the fundamental reason spreadsheets are preferred to calculators for repetitive work. (Not examinable as a proof, but frequently tested as a concept.)

3.3 Relative Referencing

Here is the observation that changes everything. Suppose column `D` contains `=B2*C2` in row 2, and we copy this formula down to rows 3, 4, 5 using the fill handle (the small square at the

bottom-right corner of the selected cell). We do not obtain `=B2*C2` everywhere. The spreadsheet adjusts the addresses automatically:

Cell	Formula after copying	Meaning
D2	=B2*C2	quantity 2 × price 2
D3	=B3*C3	quantity 3 × price 3
D4	=B4*C4	quantity 4 × price 4

DEFINITION **Relative reference**

A **relative reference** (written without any `$`, e.g. `B2`) is a reference that is adjusted automatically when the formula is copied to another cell: the column letter changes when copying horizontally, and the row number changes when copying vertically. The spreadsheet remembers the position of the referenced cell relative to the formula, not the address itself.

Intuitively, `=B2*C2` typed in `D2` really means "multiply the cell two columns to my left by the cell one column to my left". When copied to `D3`, the same instruction becomes `=B3*C3`. This is exactly what we want when the same calculation must be repeated on many rows of data — one formula, copied once, serves a thousand rows.

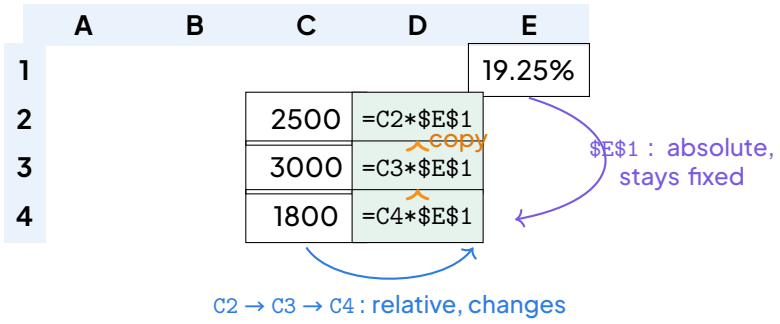
3.4 Absolute Referencing with $

Relative referencing is wonderful — until it is not. Suppose all prices must be increased by a VAT rate of 19.25% stored once in cell `E1`. In `D2` we write `=C2*E1` and copy down. In `D3` the formula becomes `=C3*E2`: the price reference correctly moved, but the VAT reference also moved to the empty cell `E2`, producing zero. We need a way to say: "this reference must **never** move".

DEFINITION **Absolute reference**

An **absolute reference** is a cell reference written with dollar signs, e.g. `$E$1`, that never changes when the formula is copied, whether vertically or horizontally. The `$` before the column letter fixes the column; the `$` before the row number fixes the row.

The correct formula in `D2` is therefore `=C2*$E$1`. Copied down, it becomes `=C3*$E$1`, `=C4*$E$1`, … The price adapts to each row; the VAT rate stays pinned to `E1`.



Copying `=C2*$E$1` down a column: the relative reference `C2` follows the row, while the absolute reference `$E$1` (the VAT rate) remains fixed.

WARNING **The classic exam trap: forgetting the $**

The most frequent mistake at the GCE is writing `=C2*E1` (relative) for a fixed rate and copying it down. Every row below the first then refers to an empty or wrong cell, giving zeros or absurd values. Rule of thumb: any value stored in a single cell and used by many formulae (a tax rate, an exchange rate, a coefficient, a pass mark) must be referenced absolutely with `$`.

METHOD **Converting a reference with the F4 key**

To change the type of a reference without typing the `$` signs by hand:

1. Double-click the cell containing the formula (or click in the formula bar).
2. Click on the reference you want to change (or place the cursor inside it).
3. Press the F4 key repeatedly to cycle through the four forms:

$$\texttt{E1} \xrightarrow{F4} \texttt{\$E\$1} \xrightarrow{F4} \texttt{E\$1} \xrightarrow{F4} \texttt{\$E1} \xrightarrow{F4} \texttt{E1}$$

4. Press Enter to validate.

(On some laptops, hold Fn while pressing F4.)

3.5 Mixed Referencing

Sometimes we want to fix only the column or only the row. This gives the two **mixed references**:

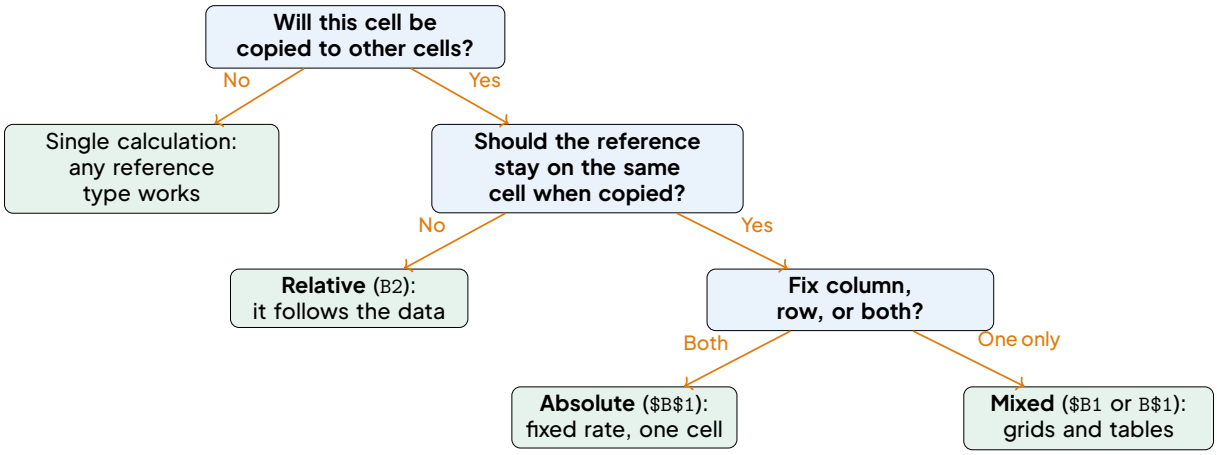
- `$B1` : the **column B is fixed**, the row is free to change;
- `B$1` : the **row 1 is fixed**, the column is free to change.

The classic application is a **multiplication table**: numbers 1 to 10 down column `A` (rows 2 to 11)

and across row 1 (columns `B` to `K`). The single formula typed in `B2` is:

```
=$A2 * B$1
```

When copied down, `$A2` keeps pointing to column `A` (its row changes), and `B$1` keeps pointing to row 1. When copied across, `$A2` stays on column `A` while `B$1` moves along the header row. One formula fills the whole 10 × 10 grid. The same technique builds a price grid (unit price × quantity) or a fees table (base fee × number of terms).



Decision guide for choosing between relative, absolute and mixed referencing.

3.6 Basic Functions: SUM, AVERAGE, MAX, MIN, COUNT, COUNTA

Adding twenty cells with `=B2+B3+B4+...` would be tedious. Spreadsheets provide built-in **functions**: named procedures applied to a range of cells. The values given to a function between its parentheses are called its **arguments**.

Function	What it computes	Example
`SUM(range)`	Total of all numbers in the range	`=SUM(B2:B20)`
`AVERAGE(range)`	Arithmetic mean of the numbers	`=AVERAGE(B2:B20)`
`MAX(range)`	Largest value in the range	`=MAX(B2:B20)`
`MIN(range)`	Smallest value in the range	`=MIN(B2:B20)`
`COUNT(range)`	Number of cells containing numbers	`=COUNT(B2:B20)`
`COUNTA(range)`	Number of non-empty cells (text included)	`=COUNTA(A2:A20)`

WARNING **COUNT versus COUNTA**

`COUNT` ignores text and empty cells; `COUNTA` counts everything that is not empty, including names. To count how many students have a mark, use `COUNT` on the marks column; to count how many students are registered, use `COUNTA` on the names column. Confusing the

two is a classic source of wrong class sizes.

AutoSum. The AutoSum button (Σ, on the Home tab) inserts `=SUM(...)` automatically, guessing the range of numbers directly above or to the left of the active cell. Always check the proposed range before pressing Enter — AutoSum stops at the first empty cell, so a gap in the data silently truncates the total. The drop-down arrow next to Σ also gives quick access to AVERAGE, MAX, MIN and COUNT.

Status bar quick calculations. When you select a range of cells containing numbers, the status bar at the bottom of the window instantly displays the **Average**, **Count** and **Sum** of the selection, without typing any formula. This is ideal for a quick check, but note that these values are not stored anywhere in the sheet.

A note on separators. Depending on the regional settings of the computer, the arguments of a function are separated either by a comma , (English settings) or by a semicolon ; (French settings): `=SUM(B2,C2)` or `=SUM(B2;C2)`. In the examination, follow the convention shown on the machine in front of you; both are accepted.

3.7 Common Error Values

When a formula cannot be evaluated, the spreadsheet displays an error value beginning with #. Reading these messages correctly is a diagnostic skill tested at the GCE.

Error	Cause	Remedy
`#DIV/0!`	Division by zero or by an empty cell, e.g. `=A2/B2` with `B2` empty	Check the denominator; test for zero before dividing
`#VALUE!`	Wrong type of operand, e.g. multiplying a number by text	Ensure all referenced cells contain numbers
`#NAME?`	Unrecognised name, usually a misspelt function (`=SOMME` in an English version, `=AVRAGE`)	Correct the spelling of the function
`#REF!`	Invalid reference: a cell used by the formula was deleted (row or column removed)	Rebuild the formula; avoid deleting referenced cells

3.8 Practical Build: A Class Mark Sheet

We now assemble everything into a realistic GCE-style task: the mark sheet of an Upper Sixth class, with marks out of 20 in three subjects.

WORKED EXAMPLE 3 **Building the mark sheet step by step**

Statement. Columns: `A` = Name, `B` = ICT, `C` = Maths, `D` = Physics (rows 2 to 6 for five students). Cell `G1` contains the number of subjects, 3. In column `E` compute each student's total, in column `F` the average, and below the table the class average, highest and lowest

averages.

SOLUTION

Solution.

1. Totals (relative references). In `E2` type `=SUM(B2:D2)` and copy down to `E6`. Each row adjusts automatically: `E3` contains `=SUM(B3:D3)`, etc.
2. Averages (absolute reference). In `F2` type `=E2/$G$1` and copy down. The total `E2` is relative (it follows the row); the divisor `$G$1` is absolute because the number of subjects lives in one single cell. Equivalently, `=AVERAGE(B2:D2)` works here, but the absolute-reference version is the one examiners look for.
3. Class statistics. In `F8`: `=AVERAGE(F2:F6)`; in `F9`: `=MAX(F2:F6)`; in `F10`: `=MIN(F2:F6)`; in `F11`: `=COUNT(F2:F6)` to verify that five averages exist.
4. Rank (preview). In `G2`: `=RANK(F2,$F$2:$F$6)` and copy down. Note again the mixed need: the student's own average `F2` is relative, but the list against which we rank must be absolute, otherwise it would slide down during the copy. (Ranking functions are studied in detail in the next section.)
5. Test the recalculation. Change one ICT mark: the total, the average, the class statistics and the ranks all update instantly — the automatic recalculation property in action.

EXERCISE 3 **Fees at a bilingual college**

A college stores in `B1` the termly fee per student (25000 FCFA) and in column `A` (rows 2 to 8) the number of children per family. (a) Write the formula in `B2` computing each family's termly bill, designed to be copied down. (b) Write the formula in `B10` giving the total collected. (c) The fee rises to 27000 FCFA: what single action updates every result?

SOLUTION **Exercise 1**

(a) `=A2*$B$1` — the family size is relative, the fee cell absolute. (b) `=SUM(B2:B8)`. (c) Simply type the new value 27000 in `B1`: every formula recalculates automatically, which is precisely why the fee was isolated in its own cell and referenced absolutely.

EXERCISE 4 **Choosing the reference type**

Cell `D1` holds an exchange rate (1 dollar = 600 FCFA). Column `B` (rows 2 to 9) holds amounts

in dollars. (a) Write the formula in C2, to be copied down, converting each amount to FCFA. (b) Predict, without a computer, what formula appears in C5 after the copy. (c) What error would appear in C3 if the rate had been referenced relatively as D1 and the cell D2 were empty?

SOLUTION **Exercise 2**

(a) =B2*$D$1. (b) =B5*$D$1: the relative part moved from row 2 to row 5, the absolute part stayed fixed. (c) With =B2*D1 copied down, C3 would contain =B3*D2; since D2 is empty it is treated as 0, so C3 would display 0 — a silent, dangerous wrong result rather than an error message.

KEY POINTS **Key points**

- Every formula begins with =; operators are + - * / {} %; priority follows BODMAS, parentheses first.
- A relative reference (B2) moves when copied; an absolute reference ($B$1) never moves; a mixed reference ($B1, B$1) fixes only the column or only the row. Use F4 to cycle through the forms.
- Fixed values used by many formulae (rates, coefficients) go in their own cell and are referenced with $.
- Master SUM, AVERAGE, MAX, MIN, COUNT, COUNTA; AutoSum and the status bar speed up the work.
- Read errors as diagnoses: #DIV/0! (division by zero), #VALUE! (wrong type), #NAME? (misspelt function), #REF! (deleted cell).
- Formulae recalculate automatically whenever a referenced cell changes.

With referencing and basic formulae mastered, we are ready for the next step: powerful logical and statistical functions, charts that make data speak, and workbooks in which several worksheets cooperate.

4 ADVANCED FUNCTIONS, CHARTS AND LINKED WORKSHEETS

In the previous section you learnt how to reference cells (relative, absolute and mixed references) and how to write basic formulae such as =B2+C2 or =SUM(B2:B10). Those formulae al-

ways do the same thing: they compute. But real-life school and business data often require the spreadsheet to make decisions, to search for information in a table, to summarise only part of the data, and to present results visually. That is exactly what this section is about: advanced functions, charts, and the linking of several worksheets into one coherent workbook.

4.1 Logical Functions: IF, Nested IF, AND, OR

Intuition. Imagine the examinations office of a school in Bamenda. A mark of 50 or more means "Admitted", otherwise "Failed". The spreadsheet must test a condition and react accordingly. This is the job of the **IF** function.

METHOD **Writing an IF function**

The general syntax is:

```
=IF(condition, value_if_true, value_if_false)
```

1. Write the **condition** using comparison operators: =, >, <, >=, <=, <> (not equal).
2. Write the result to display when the condition is TRUE.
3. Write the result to display when the condition is FALSE.
4. Text results must be enclosed in double quotes, e.g. `"Admitted"`.

EXEMPLE **Simple IF: admission decision**

Cell B2 contains a candidate's average. In C2 we write:

```
=IF(B2>=50, "Admitted", "Failed")
```

If B2 = 63, C2 displays Admitted; if B2 = 41, C2 displays Failed.

Nested IFs. When there are more than two possible outcomes, we place one IF inside another. For a grading scale (A: 80 and above, B: 60–79, C: 50–59, F: below 50):

```
=IF(B2>=80,"A", IF(B2>=60,"B", IF(B2>=50,"C","F")))
```

Read it from left to right: each FALSE branch opens a new test. Note that the number of closing brackets equals the number of IFs opened.

AND and OR. Sometimes the condition itself is compound. `AND(t1, t2, ...)` is TRUE only if all tests are true; `OR(t1, t2, ...)` is TRUE if at least one test is true. They are normally used inside IF:

```
=IF(AND(B2>=50, C2>=50), "Admitted", "Repeat")
```

admits a student only when both the written mark (B2) and the practical mark (C2) reach 50, whereas

```
=IF(OR(B2<30, C2<30), "Weak", "OK")
```

flags a student as weak as soon as one of the two marks falls below 30.

WARNING **Common mistakes with logical functions**

- Forgetting the quotes around text: `=IF(B2>=50, Admitted, Failed)` produces an error.
- Writing `80<=B2` instead of `B2>=80`: the cell reference must come first.
- In nested IFs, testing in the wrong order (e.g. testing `>=50` before `>=80`) makes the higher grades unreachable, because the first true condition stops the evaluation.

4.2 Conditional Aggregation: SUMIF, COUNTIF, AVERAGEIF

Intuition. The bursar wants the total fees paid by Form Five students only, not by the whole school. Filtering by hand is slow and error-prone. Conditional aggregation functions do it in one formula.

DEFINITION **Conditional aggregation functions**

- `=SUMIF(range, criteria, sum_range)` adds the cells of `sum_range` whose corresponding cells in `range` satisfy the `criteria`.
- `=COUNTIF(range, criteria)` counts how many cells in `range` satisfy the criteria.
- `=AVERAGEIF(range, criteria, average_range)` averages only the matching values.

The criteria can be a value (`"Form Five"`), a comparison (`">=50"`) or a cell reference.

WORKED EXAMPLE 4 **Fees paid by Form Five only**

Column A (A2:A30) holds the class of each student, column C (C2:C30) the fees paid. Write the formula giving (a) the total fees paid by Form Five students, (b) the number of Form Five students.

SOLUTION

(a) `=SUMIF(A2:A30, "Form Five", C2:C30)` — the test is applied to column A, but the amounts are taken from column C.

(b) `=COUNTIF(A2:A30, "Form Five")`.

If the class name were stored in cell E1, we would write `=SUMIF(A2:A30, E1, C2:C30)` so that changing E1 changes the class analysed.

4.3 Lookup Functions: VLOOKUP and HLOOKUP

Intuition. A shop in Douala keeps a price list: product code in column A, unit price in column B. On an invoice, the cashier types only the code, and the spreadsheet must fetch the price automatically. This is a lookup.

DEFINITION **VLOOKUP**

VLOOKUP is a function that searches a value in the first column of a range and returns a value from another column of the same row. Its syntax is:

```
=VLOOKUP(lookup_value, table_range, col_index, [range_lookup])
```

where `col_index` is the number of the column (counted from the left of the table) to return, and `range_lookup` is `FALSE` for an exact match or `TRUE` for an approximate match. **HLOOKUP** works the same way but searches the first row of a horizontal table.

EXEMPLE **Retrieving a unit price**

The price list occupies `$A$2:$B$20` (code, price). In the invoice, the code is typed in D2. The price is obtained by:

```
=VLOOKUP(D2, $A$2:$B$20, 2, FALSE)
```

If D2 contains “P07” and the row of the list whose code is “P07” shows a price of 1500, the formula returns 1500.

WARNING **The two classic VLOOKUP traps**

- Forgetting to fix the table range with $. If you write `A2:B20` and copy the formula down, the range slides to `A3:B21`, `A4:B22`, … and lookups fail. Always use `$A$2:$B$20`.
- Omitting FALSE. Without the fourth argument, VLOOKUP assumes an approximate match and may return the price of the wrong product. For codes and names, always end with `FALSE`.

4.4 Text and Rounding Functions

Spreadsheets also manipulate text. The most useful functions at this level are:

Function	Effect	Example result
`=CONCATENATE(A2," ",B2)` or `=A2&" "&B2`	Joins several text strings into one	"Amina Bello"
`=LEFT(A2,3)`	First 3 characters from the left	"UST" from "UST2024"
`=RIGHT(A2,4)`	Last 4 characters from the right	"2024" from "UST2024"
`=UPPER(A2)`	Converts text to capital letters	"AMINA"
`=ROUND(B2,2)`	Rounds a number to 2 decimal places	12.46 from 12.457

The **&** operator is a quick alternative to CONCATENATE. A typical use in a school is building a student ID: `=UPPER(LEFT(B2,3))&"-"&C2` produces e.g. "BEL-045".

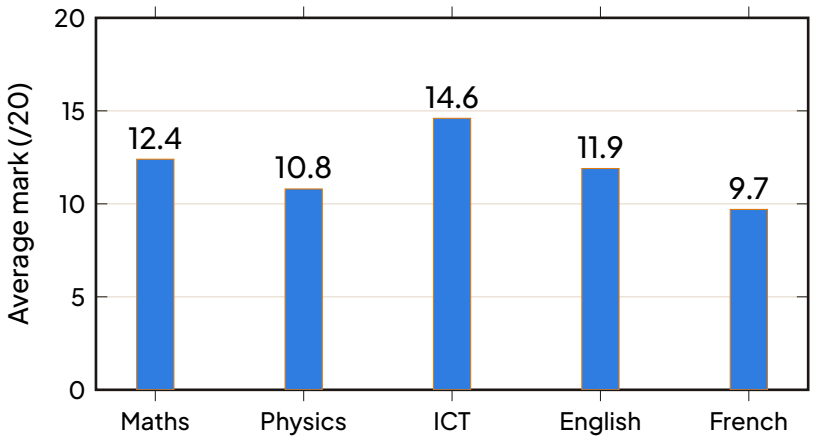
4.5 Charts: Types and Appropriate Use

A table of numbers is precise but hard to read; a **chart** reveals patterns at a glance. Choosing the right type is an examinable skill.

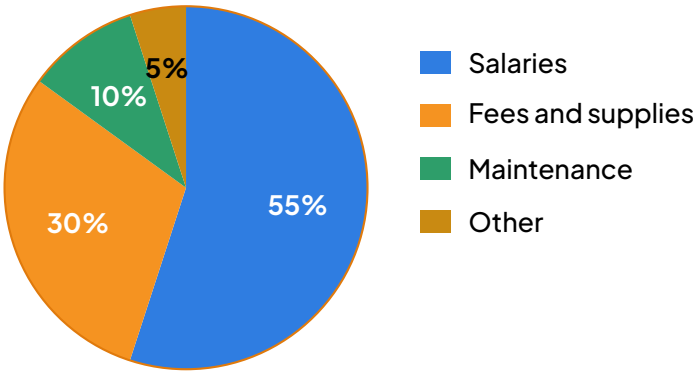
Chart type	Best for	Example
Column	Comparing values across categories	Average mark per subject
Bar	Same as column, with long category names	Enrolment per school
Line	Showing a trend over time	Fees collected per month
Pie	Showing parts of a whole (percentages)	Budget distribution

KEY POINTS **GCE exam tip — choosing the chart**

Know which chart type suits which data: pie for parts of a whole, line for trends over time, column (or bar) for comparisons between categories. Never use a pie chart to show a trend, and never use a line chart for unrelated categories.



Column chart: average marks per subject for an Upper Sixth class. ICT is immediately visible as the strongest subject.



Pie chart: distribution of a school budget. Each slice is a part of the whole (100%).

Creating and customising a chart. Select the data (including headings), then Insert → Chart and choose the type. A good chart is then customised with:

- a **chart title** stating what is shown ("Average marks per subject");
- **axis titles** with units ("Mark /20");
- a **legend** when several data series are plotted;
- **data labels** displaying the exact value on each column or slice;
- a **chart style** (colours, gridlines) that keeps the chart readable.

PROPERTY **Charts are linked to their data**

A chart created from a range is linked to that range: when the source values change, the chart updates automatically. There is no need to redraw it. (This property itself is not proved; it is verified practically and often asked as a short-answer question.)

EXPERIMENT **Verifying the automatic update**

Create a column chart from the marks of five students. Then change one mark in the table (e.g. from 10 to 18). Observe that the corresponding column grows immediately. This confirms that the chart is a view of the data, not a fixed picture.

4.6 Linking Worksheets and Consolidating Data

Intuition. A school keeps one sheet per term: `Term1`, `Term2`, `Term3`. The annual summary must combine them without retyping. The solution is the **cross-sheet reference**.

DEFINITION **Cross-sheet and cross-workbook references**

To use a cell of another worksheet in a formula, prefix the address with the sheet name and an exclamation mark:

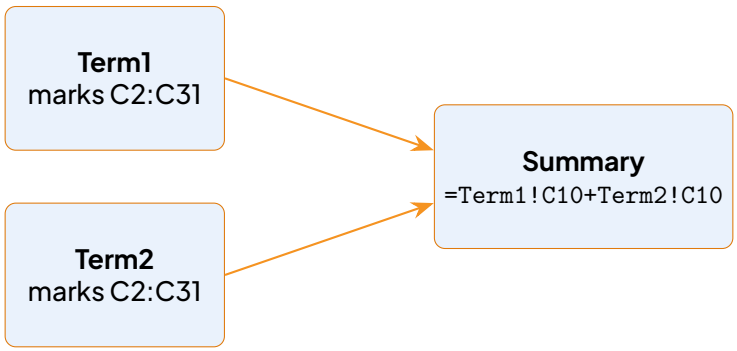
```
=Sheet2!B5      =Term1!C10
```

If the sheet name contains spaces, enclose it in single quotes: `='Term 1'!C10`. To reference another workbook, add the file name in square brackets: `=[Fees.xlsx]Sheet1!B5`.

Consolidation. The annual total of a student whose mark is in cell C10 of each term sheet is:

```
=Term1!C10 + Term2!C10 + Term3!C10   or   =SUM(Term1:Term3!C10)
```

The second form, a **3-D reference**, adds the same cell across a range of consecutive sheets — very efficient for consolidation.



Cross-sheet references: the Summary sheet pulls values from Term1 and Term2. Changing a mark in Term1 updates the Summary automatically.

Maintaining links. Links to other workbooks store the path of the source file. When the destination file opens, the spreadsheet offers to update links so that recent changes in the source are reflected.

WARNING **Broken links**

If the source workbook is moved, renamed or deleted, the link breaks: the formula keeps the last known value or returns an error such as `#REF!`. To avoid this: keep linked files in the same folder, move them together, and use Edit Links (or the equivalent command) to repair paths when necessary.

WORKED EXAMPLE 5 **Consolidating term totals**

Sheet `Term1` has total fees collected in B20, sheet `Term2` in B20, sheet `Term3` in B20. In the `Annual` sheet, write (a) the grand total, (b) the average per term.

SOLUTION

(a) `=Term1!B20 + Term2!B20 + Term3!B20`, or more compactly `=SUM(Term1:Term3!B20)`.

(b) `=AVERAGE(Term1:Term3!B20)`.

If a fee figure is corrected in Term2, both results update automatically because the Annual sheet is linked to the term sheets.

DID YOU KNOW? **Spreadsheets in Cameroonian administration**

Schools, councils and businesses across Cameroon use linked workbooks every day: a

bursary consolidates fee payments per class into one annual workbook, and mobile-money agents (MTN MoMo, Orange Money) export daily transaction sheets that are summarised monthly with SUMIF and charts. Mastering cross-sheet links is therefore a directly employable skill.

EXERCISE 5 **Practice**

A workbook contains a sheet `Products` (A: code, B: unit price) and a sheet `Sales` (A: code sold, B: quantity).

1. In Sales!C2, write the VLOOKUP formula that fetches the unit price of the code in A2.
2. In Sales!D2, write the formula for the line total, then in D20 the grand total.
3. In a sheet `Report`, write the formula counting how many sales lines have a quantity greater than 10.
4. State which chart type you would use to show the share of each product in the grand total, and justify.

SOLUTION **Exercise 1**

1. `=VLOOKUP(A2, Products!$A$2:$B$50, 2, FALSE)` — fixed range and exact match.
2. `=B2*C2` copied down; grand total `=SUM(D2:D19)`.
3. `=COUNTIF(Sales!B2:B19, ">10")`.
4. A pie chart, because each product's sales are parts of a whole (the grand total).

5 WHAT-IF ANALYSIS AND INTEGRATION ACTIVITIES

In the previous section we learnt to build powerful formulas, visualise results with charts and link several worksheets together. Our spreadsheet models are now complete: give them input values and they instantly compute results. But real life often asks the reverse question. A student does not ask "what is my average?" but rather "**what mark do I need** in Sequence 5 to reach an average of 10/20?" A bursar does not ask "what is the profit?" but "**what happens to the profit** if the price of exercise books rises to 350 FCFA?" Answering such questions is the purpose of **what-if analysis**, the last great family of spreadsheet tools. We shall then close the chapter with **integration activities** that combine Word and Excel exactly as required in the GCE practical examination.

5.1 The Concept of What-If Analysis

DEFINITION **What-if analysis**

What-if analysis is the process of changing the input values of a spreadsheet model in order to observe the effect on the results computed by formulas, without destroying the model itself.

The intuition is simple: a spreadsheet is a chain inputs → formulas → results. What-if analysis lets us play with the first link and watch the last link move. Excel offers three main tools, each suited to a different kind of question:

Tool	Question it answers	Example
Goal Seek	What single input gives a chosen result?	Mark needed for average 10/20
Scenario Manager	How do fixed sets of inputs compare?	Best / expected / worst budget
Data Table	How does the result vary over a range of inputs?	Profit for prices 300, 350, 400…

5.2 Goal Seek: Working Backwards from a Target

Consider a concrete situation. A student has sequence marks 8, 11, 9 and 12 out of 20, and the average is computed by a formula, say `=AVERAGE(B2:G2)` placed in cell `H2`, where `G2` is the (still empty) Sequence 5 mark. The student wants to know: what mark in Sequence 5 gives an average of exactly 10? One could try 9, then 10, then 11… by trial and error. **Goal Seek** automates this search.

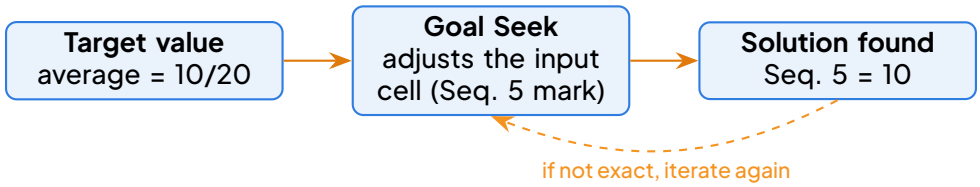
DEFINITION **Goal Seek**

Goal Seek is a what-if tool that works backwards from a desired result: you give it a target value for a formula cell, and it adjusts one input cell until the formula reaches that target.

METHOD **Using Goal Seek**

1. Click the formula cell whose result you want to control (e.g. the average `H2`).
2. Go to Data → What-If Analysis → Goal Seek.
3. Fill the three boxes: Set cell = the formula cell (`H2`); To value = the target (`10`); By changing cell = the input cell (`G2`, the Sequence 5 mark).
4. Click OK. Excel displays the solution; click OK again to keep it or Cancel to restore the

old value.



Goal Seek logic: from the desired result back to the required input.

EXEMPLE **Mark needed for the average**

With marks 8, 11, 9, 12 in `B2:E2`, an empty `G2` and `=AVERAGE(B2:G2)` in `H2`, Goal Seek with Set cell `H2`, To value `10`, By changing cell `G2` returns `G2 = 10`. Indeed (8+11+9+12+10)/5 = 50/5 = 10.

WARNING **Goal Seek traps**

The changing cell must contain a plain value (or be empty), never a formula — Goal Seek cannot overwrite a formula. Also, if no exact solution exists (e.g. asking for an average of 25/20), Goal Seek reports that it "may not have found a solution"; always check the result with your own eyes.

5.3 Scenarios: Comparing Named Sets of Values

A school bursar preparing the budget for the new academic year does not know in advance how many students will enrol or what the price of chalk will be. Rather than creating three separate files, he saves three **scenarios** in one sheet.

DEFINITION **Scenario**

A **scenario** is a named set of input values stored inside a worksheet, which can be switched on at any moment to compare the resulting outcomes (for example best case, expected case, worst case).

METHOD **Creating and viewing scenarios**

1. Build the model (e.g. `=B2*B3-B4` for profit from fees).
2. Go to Data → What-If Analysis → Scenario Manager → Add.
3. Name the scenario (`Worst case`), select the changing cells, type their values, then OK.
4. Repeat for `Expected case` and `Best case`.
5. Select a scenario and click Show to apply it; use Summary to generate a comparison report on a new sheet.

EXEMPLE **School canteen budget**

Profit = (meals sold × price) − costs. Scenarios: Worst (200 meals, 250 FCFA, costs 60 000), Expected (300 meals, 300 FCFA, costs 70 000), Best (400 meals, 300 FCFA, costs 75 000). Switching scenarios instantly shows profits of −10 000, 20 000 and 45 000 FCFA respectively — the bursar sees at a glance that the canteen must sell at least about 240 meals to break even.

5.4 Data Tables: Varying One or Two Inputs Systematically

Goal Seek varies one input to hit one target; scenarios compare a few fixed cases. But sometimes we want a whole map of results: profit for every price from 250 to 500 FCFA, or for every combination of price and quantity. This is the job of **data tables**.

DEFINITION **Data table**

A **data table** is a range of cells in which Excel automatically re-evaluates a formula for a whole list of input values. A one-variable data table shows how results change with one input; a two-variable data table shows results for every pair of values of two inputs, arranged as a grid.

METHOD **Building a one-variable data table**

1. Build the model, e.g. profit `=B1*(B2-B3)` where `B1` is quantity, `B2` price, `B3` unit cost.
2. In a column, list the trial prices (250, 300, 350, 400); in the cell above and one column to the right, reference the result formula (e.g. `=B4` if the profit is in `B4`).
3. Select the whole block, then Data → What-If Analysis → Data Table; set the Column input cell to `B2` and click OK.

For a two-variable table, place prices down the left column, quantities across the top row, the formula reference in the top-left corner cell, then fill both a Row input cell and a Column input cell.

EXEMPLE **Profit grid for the co-operative**

A two-variable data table with prices 300–500 FCFA (rows) and quantities 100–400 (columns) produces a grid of profits. The committee immediately sees which (price, quantity) pairs keep the co-operative profitable, without typing a single extra formula.

WARNING **Choosing the right tool**

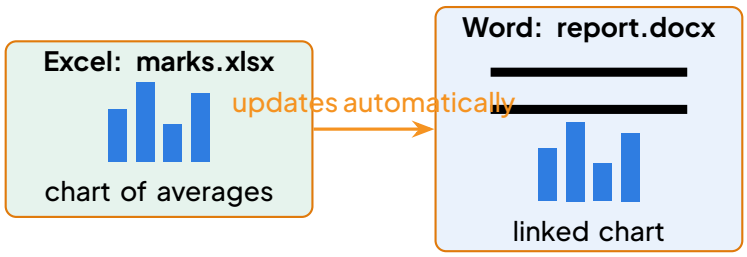
Exam questions often test this choice. Remember: one input, one exact target → Goal Seek; a few named cases to compare → Scenarios; a whole range of values to explore → Data table. Using Goal Seek to fill a table of 20 values is a classic waste of exam time.

5.5 Integration Activity 1: A Complete School Report

The GCE practical paper rarely tests Word or Excel in isolation; it asks you to combine them. The first integration activity is a school report: an Excel mark sheet (formulas, ranking, chart) presented inside a Word document.

METHOD **Producing the combined report**

1. In Excel, build the mark sheet: totals with `SUM`, averages with `AVERAGE`, ranks with `RANK`, decisions with `IF`, and a bar chart of subject averages.
2. In Word, type the report text (heading with styles, introduction, comments).
3. Select the Excel chart, copy it (`Ctrl+C`), then in Word use Home → Paste → Paste Special → Paste Link → Microsoft Excel Chart Object.
4. Save both files in the same folder. When the Excel marks are corrected, the chart in Word updates automatically.



A chart linked from Excel into a Word report: editing the workbook refreshes the document.

5.6 Object Linking and Embedding (OLE)

Paste Special offers three behaviours that the GCE loves to ask about. Together they form the technology called **OLE** (**O**bject **L**inking and **E**mbedding).

Mode	Behaviour	When to use it
Simple paste (picture/table)	A static copy; no connection with Excel	Final document, data will never change
Linking	Word stores a reference to the Excel file; the chart updates when the workbook changes	Both files stay together and data evolves
Embedding	A full copy of the Excel object is stored inside the Word file; double-clicking edits it with Excel tools	The Word file must travel alone (e-mail, USB key)

WARNING **The broken-link trap**

A very common mistake: a candidate links a chart into Word, then deletes or moves the Excel file (or copies only the `.docx` to a USB key). The link breaks and the chart can no longer update — it may even disappear. Rule of thumb: if the Excel file will not travel with the document, embed the chart instead of linking it.

5.7 Integration Activity 2: Mail-Merging Result Slips

Sending result slips to 200 parents does not mean typing 200 letters. **Mail merge** combines one Word template with the Excel mark sheet used as a **data source**.

METHOD **Mail merge of result slips**

1. In Excel, prepare a clean table: first row = field names (`Name`, `Class`, `Average`, `Rank`), one

row per student. Save and close the file.

2. In Word: Mailings → Start Mail Merge → Letters.
3. Select Recipients → Use an Existing List → choose the Excel file and the correct worksheet.
4. Place the cursor in the letter and use Insert Merge Field to add `<<Name>>`, `<<Average>>`, `<<Rank>>`.
5. Preview Results to check a few records, then Finish & Merge → Print Documents or Edit Individual Documents.

WORKED EXAMPLE 6 **Choosing and applying a what-if tool**

Statement. A candidate's model computes the term average in `H2` from five sequence marks. (a) Which tool finds the Sequence 5 mark needed for an average of exactly 10/20? (b) Which tool compares the budget outcomes for enrolments of 400, 500 and 600 students? (c) Which tool shows the profit for every ticket price from 100 to 1000 FCFA in steps of 100? Justify each answer.

SOLUTION

Solution. (a) Goal Seek: a single input (one mark) must be adjusted to reach one exact target (10/20) — set cell `H2`, to value `10`, by changing the Sequence 5 cell. (b) Scenario Manager: three named, fixed sets of inputs (400/500/600 students) are stored and switched to compare outcomes. (c) A one-variable data table: the result must be evaluated over a whole range of values of a single input (the price), which is exactly what a data table automates.

5.8 Revision Synthesis and GCE Practice

KEY POINTS **Chapter synthesis**

- Word processing: styles, headers/footers, tables of contents, mail merge produce professional documents.
- Spreadsheets: formatting, sorting, filtering, absolute referencing (`$B$2`), functions (`SUM`, `AVERAGE`, `IF`, `VLOOKUP`, `RANK`), charts and linked sheets turn raw data into information.
- What-if analysis: Goal Seek (one input, one target), Scenarios (named cases), Data tables (ranges of one or two inputs).

- OLE: paste (static), link (updates, needs the source file), embed (portable, larger file).

EXERCISE 6 **GCE-style practice**

A school stores CA marks in `Sheet1` and exam marks in `Sheet2`. (a) Write the formula computing the term total on `Sheet1` that pulls the exam mark from `Sheet2`. (b) State the what-if tool and the exact dialog entries to find the exam mark giving a term average of 10/20. (c) The principal wants result slips for all parents: describe the mail-merge steps. (d) The report chart must remain correct even if the Excel file is deleted: should you link or embed? Justify.

SOLUTION **Exercise 1**

(a) `=B2+Sheet2!B2` (with appropriate cells). (b) Goal Seek: Set cell = average cell, To value = `10`, By changing cell = the exam mark cell on `Sheet2`. (c) Mailings → Start Mail Merge → Letters; Select Recipients → Use Existing List → the Excel sheet; Insert Merge Fields; Preview; Finish & Merge. (d) Embed: linking would break when the source file is deleted, whereas embedding stores a full copy of the object inside the Word document.

DID YOU KNOW? **GCE exam tip**

Practical papers very often require the full chain: a formula sheet in Excel, a chart, a linked or embedded object in a Word report, and a merged letter — all within a limited time. Practise this complete workflow at home until you can do it in under 45 minutes; on exam day, save your files early and often, and never delete a source file before checking every linked object.

6 INTEGRATION ACTIVITY

6.1 Aims of the activity

This integration activity closes the chapter by mobilising, in one realistic project, all the competences studied in Lessons 58 to 63: word-processing features for producing documents, formatting and managing spreadsheet data, cell referencing and formulae, advanced functions and charts, linking worksheets, and what-if analysis. By the end of the activity, each pair of learners should be able to:

- structure a multi-sheet workbook (one sheet per sequence plus a summary sheet) and link sheets with **cross-sheet references**;
- use **relative**, **absolute** and **mixed references** correctly when copying formulae;
- compute totals, averages and ranks, and apply logical and statistical functions (`IF`, `SUMIF`, `COUNTIF`, `AVERAGE`, `RANK`, `MAX`, `MIN`);
- build and format a **bar chart** and a **pie chart** from spreadsheet data;
- use **Goal Seek** to answer a what-if question;
- produce a formal Word document with **heading styles**, an **automatic table of contents** and a linked chart;
- perform a **mail merge** using an Excel sheet as data source;
- demonstrate that the package is automated: changing one raw mark updates everything.

KEY POINTS **Success criteria**

The package is successful if (i) no result is typed twice — everything is computed by formulae or links; (ii) the charts and the Word report update when a mark changes; (iii) the mail merge produces 20 personalised result slips without retyping any name or mark.

6.2 Situation

EXPERIMENT **End-of-term reports at GHS Nkwen**

Government High School Nkwen (Bamenda, North-West Region) still prepares end-of-term reports by hand. The Principal, Mrs. A. Mbuh, has asked the Upper Sixth ICT class to digitalise the process for one pilot class, Form 5A, before generalising it to the whole school.

The class teacher gives you the raw marks (out of 20) of 20 students in 6 subjects for each of the two sequences of the term. The subjects and their coefficients are those of the Anglophone sub-system for Form 5: English Language (coef. 3), Mathematics (coef. 4),

Physics (coef. 3), Chemistry (coef. 2), Biology (coef. 2), Computer Science (coef. 2). The total coefficient is 16; the weighted term average is computed out of 20. The school rule for the term decision is:

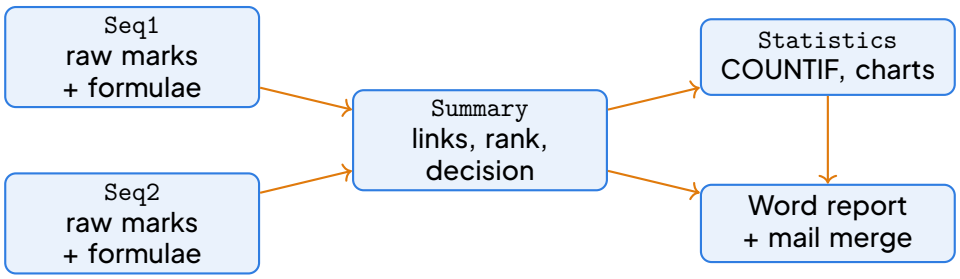
Weighted average	Decision
Average ≥ 10/20	Admitted (passes to the next sequence in good standing)
Average < 10/20	Repeat work / warned (parent summoned)

Extract of the data for Sequence 1 (the full list of 20 students is provided by the teacher as a CSV file exported from the school register):

Name	ENG	MAT	PHY	CHE	BIO	CSC
Abongwa Claris	12	14	11	13	15	16
Bih Terence	9	7	8	10	11	14
Che Mildred	15	16	14	12	13	17
Fru Collins	6	5	7	8	9	12
…16 other students …						

Sequence 2 marks are given in a second CSV file with the same structure. One case worries the class teacher: Bih Terence has a Sequence 2 weighted average of 9.6/20. The Principal wants to know: "What mark does Terence need in Sequence 3 Mathematics for his cumulative average to reach exactly 10/20?"
The Principal expects from each pair a complete package:

1. an Excel workbook `Form5A_Term1.xlsx` with sheets `Seq1`, `Seq2`, `Summary` and `Statistics`;
2. a one-page Word report to the Principal, with a table of contents and the chart of subject averages;
3. 20 individual result slips for parents, produced by mail merge;
4. a 5-minute presentation, ending with a live demonstration: the teacher changes one mark, and the whole package (averages, ranks, decision, charts) recalculates automatically.



Data flow of the package: raw marks are entered once; every other result is a formula or a link.

6.3 Tasks

EXERCISE 7 **Task 1 — Building the workbook**

Import the two CSV files into sheets `Seq1` and `Seq2`. Add columns Weighted total and Weighted average. Write the formula for the weighted total of the first student using the coefficients stored in a header row with mixed references (e.g. `=B2*B$1+C2*C$1+...`), then copy it down for the 20 students. Explain why the dollar sign is placed before the row number and not before the column letter.

EXERCISE 8 **Task 2 — The linked summary sheet**

On the `Summary` sheet, retrieve each student's two sequence averages using **cross-sheet references** (e.g. `=Seq1!I2`), compute the term average (mean of the two sequences), the rank with `RANK`, and the decision with `IF`. Why must the range of averages inside `RANK` be an absolute reference?

EXERCISE 9 **Task 3 — Class statistics**

On the `Statistics` sheet, compute: the number of students (`COUNTA`), the class average (`AVERAGE`), the highest and lowest averages (`MAX`, `MIN`), the number of admitted students (`COUNTIF`) and the total of averages above 10 (`SUMIF`). Compute the success rate as a percentage.

EXERCISE 10 **Task 4 — Charts**

Create (a) a bar chart of the six subject averages for the term, and (b) a pie chart showing the distribution Admitted / Repeat work. Give each chart a title, axis titles (bar chart) and

data labels in percentage (pie chart).

EXERCISE 11 **Task 5 — Goal Seek**

Using **Goal Seek** (Data → What-If Analysis), determine the mark Terence needs in Sequence 3 Mathematics for his cumulative weighted average to equal exactly 10/20. Set up the calculation so that the cumulative average is a formula depending on the unknown mark.

EXERCISE 12 **Task 6 — Word report and mail merge**

(a) Write the one-page report to the Principal using Heading 1 / Heading 2 styles, insert an automatic table of contents, and paste the bar chart as a linked object. (b) Prepare the result slip (student name, subject averages, term average, rank, decision) and run a mail merge with the `Summary` sheet as data source to produce the 20 slips.

EXERCISE 13 **Task 7 — Live demonstration**

Present the package in 5 minutes. Change Fru Collins' Sequence 2 Mathematics mark from 5 to 12 and show that the summary, statistics, charts and decision update without any retyping.

6.4 Model answer

Task 1. After import, row 1 holds the coefficients in `B1:G1` (3, 4, 3, 2, 2, 2) and `H1` contains the total coefficient 16. In `H2`:

```
=B2*B$1+C2*C$1+D2*D$1+E2*E$1+F2*F$1+G2*G$1
```

and in `I2`: `=H2/$H$1`. The reference `B$1` is mixed: the column letter stays free so that the formula reads each subject's mark and coefficient along the row, while the row is locked with `$1` so that copying down to rows 3–21 still points to the coefficient row. For Abongwa Claris:

$$\begin{aligned}\text{Total} &= 12\times 3 + 14\times 4 + 11\times 3 + 13\times 2 + 15\times 2 + 16\times 2\\ &= 36 + 56 + 33 + 26 + 30 + 32 = 213,\end{aligned}$$

giving a weighted average of 213/16 = 13.31/20 (rounded to two decimal places).

Task 2. On `Summary`, `B2` = `=Seq1!I2`, `C2` = `=Seq2!I2`, `D2` = `=AVERAGE(B2:C2)`. Rank in `E2`: `=RANK(D2,$D$2:$D$21)` — the range must be absolute (`$D$2:$D$21`) so that it does not shift when the formula is copied down; otherwise student 3 would be ranked against `D3:D22`, which is wrong. Decision in `F2`: `=IF(D2>=10,"Admitted","Repeat work")`.

Task 3. Typical formulae on `Statistics`:

Statistic	Formula
Number of students	`=COUNTA(Summary!A2:A21)`
Class average	`=AVERAGE(Summary!D2:D21)`
Highest / lowest average	`=MAX(Summary!D2:D21)` / `=MIN(Summary!D2:D21)`
Number admitted	`=COUNTIF(Summary!F2:F21,"Admitted")`
Sum of averages ≥ 10	`=SUMIF(Summary!D2:D21,">=10")`
Success rate	`=COUNTIF(Summary!F2:F21,"Admitted")/20` formatted as %

Task 4. The subject averages are computed with `=AVERAGE(Seq1!B2:B21)` etc. on the `Statistics` sheet, then selected and inserted as a **clustered bar chart** titled "Form 5A — Subject averages, Term 1", with axis titles "Subject" and "Average (/20)". The pie chart uses the two `COUNTIF` results (Admitted / Repeat work) with percentage data labels.

Task 5. First build the model. In a free area, enter Terence's estimated Sequence 3 marks for the five other subjects, and a trial mark (e.g. 10) for Mathematics in a single cell, say `K2`. Compute his Sequence 3 weighted total with the same coefficient formula (Mathematics contributes 4 × `K2`), then his Sequence 3 average, then the cumulative average `=(Seq1avg+Seq2avg+Seq3avg)/3`. Open **Data → What-If Analysis → Goal Seek**: Set cell = cumulative average cell, To value = 10, By changing cell = `K2`. Goal Seek iterates and returns the required Mathematics mark.

Numerical illustration. Suppose Terence's Sequence 1 average is 9.0/20 and his estimated Sequence 3 marks (without Mathematics) give a partial weighted total of 130. To reach a cumulative average of 10 he needs a Sequence 3 average of 3 × 10 − 9.0 − 9.6 = 11.4, i.e. a weighted total of 11.4 × 16 = 182.4, hence a Mathematics mark of (182.4 − 130)/4 = 13.1/20. Your own result will depend on his actual Sequence 1 average and estimated marks — that is precisely why Goal Seek is used rather than mental arithmetic. If Goal Seek returns a mark above 20, the honest conclusion to report is that the target of 10/20 is unreachable through Mathematics alone.

Task 6. The Word report uses **Heading 1** for "Form 5A — End-of-Term Report" and **Heading 2** for "Results overview", "Statistics" and "Recommendations"; the table of contents is inserted via **References → Table of Contents** and refreshed with F9. The chart is pasted with **Paste Special → Paste link** so it updates with the workbook. For the slips: **Mailings → Start Mail Merge → Letters**, **Select Recipients → Use an Existing List** pointing to `Form5A_Term1.xlsx`, sheet `Summary`; merge fields `Name`, `Average`, `Rank`, `Decision` are inserted, then **Finish & Merge** generates the 20 personalised slips.

Task 7. Changing Fru Collins' mark in `Seq2` propagates instantly: his average, rank and decision change, the `COUNTIF` statistics and success rate update, and both charts are redrawn — proving the package is fully automated.

WARNING **Common mistakes to avoid**

- Typing computed values by hand instead of formulae — the live demonstration then fails immediately.
- Forgetting the `$` in the `RANK` range or in the coefficient row: copied formulae then give

wrong results.

- Averaging percentages or ranks instead of raw weighted totals.
- Dividing by the number of subjects (6) instead of the total coefficient (16) when computing a weighted average.
- Pasting the chart as a simple picture instead of a linked object: the Word report no longer updates.

DID YOU KNOW? **Why this matters in Cameroon**

Schools that digitalise their reporting save days of manual calculation each term and reduce transcription errors. The same skills — linked sheets, mail merge, Goal Seek — are used daily in Cameroonian banks, mobile-money agent accounting (MTN MoMo, Orange Money) and the e-government services promoted by ANTIC. Mastering them makes you immediately useful in any administration.

7 METHODS AND WORKED EXAMPLES

8 USING FORMATTING, EDITING FEATURES OF WORD PROCESSORS FOR PRODUCING DOCUMENTS

DID YOU KNOW? **Why this chapter matters for the GCE**

The GCE Advanced Level ICT practical paper (Paper 3) and the theory paper (Paper 1) both test your ability to produce professional documents and analyse data in spreadsheets. This chapter chains the seven lessons of the official Second Term progression (Lessons 58–64) into a single workflow: from a blank page to a structured report, from raw data to a chart, from a formula to a what-if scenario, and finally to an integrated document that could be submitted to a principal, a bank manager, or an ANTIC inspector. Master every method below; they are the exact skills the examiner expects.

8.1 Producing Documents with Word Processing Features

EXPERIMENT **Observing a poorly formatted document**

Open the file `BadReport.docx` (provided in your resources). Notice: the title is centred by pressing `Space` repeatedly; headings are just bold, large text; the table of contents was typed by hand; page numbers are missing; the table columns do not align. Try to insert a new section in the middle — everything shifts. This is what not to do. The professional approach uses **styles**, **page breaks**, and **automatic fields**.

METHOD **Producing a professional document in a word processor**

To produce a well-structured document (report, letter, CV, minutes) with Microsoft Word or LibreOffice Writer:

1. Plan the structure on paper first: title page, table of contents, headings (level 1, level 2), body paragraphs, tables, images, references, appendices.
2. Set up the page: Layout → Size: A4; Margins: Normal (2.54 cm); Orientation: Portrait (or Landscape for wide tables).
3. Apply styles, not manual formatting: select each heading and apply Heading 1, Heading 2, etc.; apply Normal to body text. This enables automatic **table of contents**, **nav-**

igation pane, and consistent reformatting.

4. Format text properly: font (Calibri/Arial 11 or 12), alignment (justified for body, centred for title), line spacing (1.15 or 1.5), bullets/numbering via the Home tab.
5. Insert objects correctly:
   - Tables: Insert → Table → choose rows/columns → type data → Table Design for header row shading.
   - Pictures: Insert → Pictures → wrap text In Line with Text or Tight.
   - Page numbers: Insert → Page Number → Bottom of Page → Plain Number 2.
   - Headers/Footers: Insert → Header/Footer → type name, date, document code.
   - Footnotes/Endnotes: References → Insert Footnote.
   - Page breaks: `Ctrl+Enter` (never multiple `Enter` keys).
6. Use editing features: Find & Replace (`Ctrl+H`), Spelling & Grammar (Review → Spelling), Track Changes for collaboration.
7. Generate automatic table of contents: place cursor where TOC should appear → References → Table of Contents → Automatic Table 1.
8. Save and export: `Ctrl+S` regularly; final delivery: File → Export → Create PDF/XPS Document.

Reflexes: Never press `Enter` repeatedly to create vertical space — use Spacing Before/After in Paragraph settings. Always use styles for headings. Save every 5 minutes (`Ctrl+S`).

KEY POINTS **Word processor essentials for GCE**

- Styles = structure + automation (TOC, cross-references).
- Page break (`Ctrl+Enter`) = clean new page, immune to editing above.
- Section breaks = different headers/footers, orientation, or columns in the same document.
- Paste Special → Paste Link = live link to spreadsheet data (updates when source changes).
- PDF export = fixed layout, non-editable, standard for submission.

WARNING **Classic GCE traps in word processing**

- Manual formatting (bold + large font) instead of Heading 1 style → TOC stays empty.
- Multiple Enters for spacing → pagination breaks when text is edited.
- Typing page numbers by hand → they don't update.
- Forgetting to update fields after editing: select all (`Ctrl+A`) → `F9` to refresh TOC, page numbers, cross-references.
- Not saving as .docx/.odt before PDF → you lose the editable source.

WORKED EXAMPLE 7 **Formatting a school report — GCE style**

A form teacher asks you to produce the end-of-term report of Government High School Mfoundi. The document must have: a title page, an automatic table of contents, two sections with headings, a table of results, page numbers in the footer, and must be exported as PDF. Describe the steps you would follow in a word processor.

SOLUTION

Solution.

1. Open the word processor and set the page: Layout → Size: A4; Margins: Normal (2.54 cm); Orientation: Portrait.
2. Title page: type the school name (`Government High School Mfoundi`), the report title (`End-of-Term Report`), and the term (`Second Term 2024/2025`); centre the text, increase font size (e.g. 24 pt for school name, 18 pt for title), then insert a page break (`Ctrl+Enter`) so the next content starts on a new page.
3. On page 2, leave space for the table of contents (to be generated later). Insert a page break after the TOC placeholder.
4. Type the body of the report. Select each main section title (e.g. `1. Academic Performance`, `2. Discipline`) and apply the style Heading 1; apply Heading 2 to subsections (e.g. `1.1 Class Averages`). Apply the Normal style to paragraphs, with justified alignment and 1.15 line spacing.
5. Insert the results table: Insert → Table, choose the number of rows and columns (e.g. 6 rows × 5 columns for 5 classes + header), type the data, then format the header row (bold, shading, centred).
6. Insert page numbers: Insert → Page Number → Bottom of Page → Plain Number 2.
7. Generate the table of contents: place the cursor on page 2 (the TOC placeholder),

then References → Table of Contents → Automatic Table 1. The headings styled Heading 1/2 appear automatically with their page numbers.

8. Run the spelling checker (Review → Spelling & Grammar), save the file as `GHS_Mfoundi_Report_Term2.docx`, then export: File → Export → Create PDF/XPS Document → `GHS_Mfoundi_Report_Term2.pdf`.

Key point: the automatic table of contents works only because heading styles were used — manual bold text would not be detected. Always update the TOC (`F9`) after any edit.

8.2 Formatting and Managing Spreadsheet Data

EXPERIMENT **Observing a messy marksheet**

Open `MessyMarks.xlsx`. Column A has "Name", B has "Class", C has "Mark/20", but some cells contain "15/20", others "15", one has "fifteen". Column D "Coefficient" has "2" and "two". The total row uses `=C2+C3+...` instead of `SUM`. Try to sort by mark — the header row gets mixed in. Try to filter "Upper Sixth A" — nothing happens because the data isn't recognised as a table. This is why formatting and managing come before formulae.

METHOD **Formatting and managing data in a spreadsheet**

To present and manage data in Excel or LibreOffice Calc:

1. Enter data as a structured table: one field per column, one record per row; first row = column headers (unique, no blanks). Example: `Name | Class | Mark (/20) | Coefficient | Weighted Mark`.
2. Format cells for meaning, not just looks:
   - Number format: Home → Number Format → Currency (FCFA), Percentage, Date, Number (0 decimals for marks).
   - Never type units inside numeric cells ("25000 FCFA" becomes text). Put the unit in the column header and format the cell as Currency (symbol FCFA).
   - Fonts, borders, fill colours for header row; alignment (centre for numbers, left for text); Wrap Text for long headers; Merge & Center only for report titles above the table.
3. Adjust layout: double-click column boundary to auto-fit; View → Freeze Panes → Freeze Top Row so headers stay visible when scrolling.
4. Convert to Excel Table (`Ctrl+T`): enables structured references, auto-expansion, filter buttons on every header.

5. Sort data: click any cell in the table → Data → Sort → choose key column (e.g. `Mark`) and order (Largest to Smallest). Always include the header row in the selection.
6. Filter data: Data → Filter (or `Ctrl+Shift+L`) → drop-down arrows appear → tick/untick values (e.g. only `Upper Sixth A`). Hidden rows are not deleted.
7. Validate entries (optional but examined): Data → Data Validation → Allow: List → Source: `Pass,Fail` → creates a drop-down in the cell, preventing typos.

Reflexes: One header row, no blank rows/columns inside the data, numeric cells contain only numbers, use `Ctrl+T` for a managed table.

KEY POINTS **Spreadsheet data management**

- Raw data → Format as Table (`Ctrl+T`) → Sort/Filter → Analyse.
- Freeze Panes keeps headers visible; essential for lists > 20 rows.
- Data Validation reduces input errors (examinable: "restrict entry to a list").
- Sort rearranges rows; Filter hides rows. Both keep records intact.

WARNING **Spreadsheet data traps**

- Text in numeric column ("15/20", "Absent") → `SUM`, `AVERAGE` ignore it → wrong totals.
- Sorting only one column → data corruption (names no longer match marks). Always select the whole table.
- Merged cells inside the data area → break sorting, filtering, formulas. Merge only for titles above the table.
- Blank rows inside data → filter/sort stops at the blank.

WORKED EXAMPLE 8 **Managing a marks list — GCE style**

The marks of 5 candidates in ICT (out of 20) are entered in a worksheet: A1:E1 contain the headers `Name`, `Class`, `Mark (/20)`, `Coefficient`, `Weighted`; rows 2 to 6 contain the records. (a) Explain how to display only candidates of class "Upper Sixth A". (b) Explain how to sort the candidates by decreasing mark. (c) The teacher typed "15/20" in a Mark cell and the SUM function returned a wrong total. Explain why and correct it.

SOLUTION

Solution.

(a) Click any cell in the table, then Data → Filter (AutoFilter) or press `Ctrl+Shift+L`. Drop-down arrows appear on each header. Click the arrow of the `Class` column, untick "(Select All)", tick only `Upper Sixth A`, and click `OK`. Only the matching rows remain visible; the others are hidden, not deleted. The status bar shows "3 of 5 records found".

(b) Select the whole table (headers included, e.g. A1:E6), then Data → Sort. In the dialog: "Sort by: Mark (/20)", "Sort On: Cell Values", "Order: Largest to Smallest". Ensure "My data has headers" is ticked. Click `OK`. The rows are rearranged while each record stays intact.

(c) "15/20" is interpreted either as text or as a date (15/20 is not a valid date, so it stays text). Text is ignored by `SUM`, hence the wrong total. Correction: type only the number `15` in the cell, and indicate "out of 20" in the column header, e.g. `Mark (/20)`. Format the column as Number with 0 decimals. The sum is then correct.

Key point: sorting acts on entire rows; filtering only hides rows; numeric cells must contain numbers only.

8.3 Cell Referencing and Basic Formulae

EXPERIMENT **The $ sign mystery**

In a sheet, B1 = 10. In D1 you type `=B1*2` and drag right to E1, F1. D1=20, E1=0 (because C1 is empty), F1=0. Now type `=$B$1*2` in D1 and drag right: D1=20, E1=20, F1=20. The $ locked the reference. This is the single most examined concept in GCE spreadsheet questions.

METHOD **Using cell references and basic formulae**

1. Every formula begins with the sign **=**.
2. Relative reference (e.g. `B2`): changes when the formula is copied (row and column both shift).
3. Absolute reference (e.g. `$B$2`): fixed when copied; the $ locks the column and the row.
4. Mixed references: `$B2` (column locked, row relative), `B$2` (row locked, column relative). Press `F4` (Windows) or `Cmd+T` (Mac) to cycle: `B2` → `$B$2` → `B$2` → `$B2` → `B2`.

5. Basic functions (all take a range):
   - `=SUM(range)` — total
   - `=AVERAGE(range)` — arithmetic mean
   - `=MAX(range)` — highest value
   - `=MIN(range)` — lowest value
   - `=COUNT(range)` — count of numeric cells
   - `=COUNTA(range)` — count of non-empty cells (text or numbers)
6. To copy a formula, drag the **fill handle** (small square at bottom-right of active cell) or double-click it to auto-fill down a column.

Reflex: any value that must stay fixed in a copied formula (a coefficient, a rate, a single total, a tax percentage) must be referenced with $.

KEY POINTS **Reference types — decision guide**

Reference	Behaviour when copied	Typical use
`B2`	Both row and column change	Normal data cells (marks, quantities)
`$B$2`	Neither changes	Constants: tax rate, exchange rate, total coefficient
`B$2`	Row fixed, column changes	Row headers used across columns
`$B2`	Column fixed, row changes	Column headers used down rows

WARNING **Formula traps in GCE**

- Forgetting = → formula displays as text.
- Using comma instead of semicolon (or vice versa) depending on regional settings. Cameroon GCE uses semicolon (;) as list separator in functions: `=IF(A1>10;"Pass";"Fail")`.
- Relative reference where absolute needed → copied formula points to wrong cell (classic: dividing by a moving total).
- Circular reference (formula refers to its own cell) → error or 0.
- `COUNT` vs `COUNTA`: `COUNT` counts numbers only; `COUNTA` counts everything non-blank.

WORKED EXAMPLE 9 **Weighted average with absolute referencing**

In a worksheet, B2:B6 contain the marks of 5 students, C2:C6 the coefficients, and cell E1 contains the total of coefficients (computed by `=SUM(C2:C6)`). (a) Write the formula in D2 computing the weighted mark (mark × coefficient). (b) Write the formula in D7 computing the weighted average of the class. (c) Explain the role of the $ signs if you used `$E$1`.

SOLUTION

Solution.

(a) In D2: `=B2*C2`. Both references are relative: when the formula is dragged down to D6, it becomes `=B3*C3`, `=B4*C4`, etc., which is exactly what we want (each student's own mark × own coefficient).

(b) The weighted average is

$$\bar{m} = \frac{\sum m_i c_i}{\sum c_i}$$

so in D7: `=SUM(D2:D6)/SUM(C2:C6)`, or equivalently `=SUM(D2:D6)/$E$1` if the total of coefficients is stored in E1.

(c) If the formula `=SUM(D2:D6)/E1` were copied elsewhere (e.g. to the right), the reference E1 would move (F1, G1, …) and give errors or wrong results; writing `$E$1` locks both the column and the row so the divisor always points to the same cell E1, wherever the formula is copied.

Key point: relative references "follow" the copy; absolute references with $ stay fixed. Choosing correctly is a classic GCE question.

8.4 Advanced Functions, Charts and Linked Worksheets

EXPERIMENT **From "IF" to a dashboard**

A school bursar in Buea wants a sheet that automatically shows "Scholarship" if average ≥ 14, "Probation" if average < 10, else "Normal". Then a chart of averages per class. Then the same sheet linked to a master workbook. This lesson chains `IF`, `VLOOKUP`, charts, and cross-sheet references — exactly the "advanced" skills of Lessons 61–62.

METHOD **Logical, lookup functions and charts**

1. IF function (single condition): `=IF(logical_test; value_if_true; value_if_false)`. Example: `=IF(B2>=14;"Scholarship";IF(B2<10;"Probation";"Normal"))` (nested IF).

2. Conditional counting/summing:
   - `=COUNTIF(range; criterion)` — e.g. `=COUNTIF(C2:C50;">=14")`
   - `=SUMIF(range; criterion; sum_range)` — e.g. sum fees for class "A": `=SUMIF(B2:B50;"A";D2:D50)`
3. Lookup functions:
   - `=VLOOKUP(lookup_value; table_array; col_index_num; FALSE)` — searches first column of table, returns value from `col_index_num`. `FALSE` = exact match (required for codes, names).
   - `=HLOOKUP(...)` — horizontal lookup (rare in GCE).
4. Charts — select data with headers → Insert → Chart:
   - Column/Bar chart: compare categories (e.g. marks per student, sales per agent).
   - Line chart: show evolution over ordered time (days, months, terms).
   - Pie chart: show proportions of a whole (e.g. budget allocation, gender distribution). Only one data series.
   - Scatter (XY) chart: relationship between two numeric variables (e.g. study hours vs mark).

   Always add: Chart Title, Axis Titles, Legend (if multiple series), Data Labels (if few points).

Reflex: text criteria in formulas are written between double quotes (`"Pass"`); numeric criteria without quotes (`>10`); cell references without quotes (`B2`).

KEY POINTS **Chart choice — examiner's checklist**

Question asks for…	Choose	Why
"Compare values across categories"	Column / Bar	Length encodes magnitude
"Show trend / evolution over time"	Line	Continuity emphasises change
"Show percentage share / proportion"	Pie / Doughnut	Angle encodes part-of-whole
"Show relationship between two measures"	Scatter (XY)	Both axes numeric

WARNING **Advanced function traps**

- `VLOOKUP` without `FALSE` (or `0`) → approximate match → wrong results if first column not sorted.
- `VLOOKUP` column index counted from start of table_array, not from column A.
- Nested IFs: close every parenthesis; maximum 64 levels (but 3–4 is typical in GCE).
- `COUNTIF` criterion syntax: `">=10"` (quotes), `B2` (no quotes), `"Pass"` (quotes).
- Charting the wrong range (e.g. including the total row) → distorted chart.

WORKED EXAMPLE 10 **Decision function and chart for a canteen**

A school canteen in Bamenda records daily sales of meals in B2:B8 (Monday to Sunday). (a) Write the formula in C2 displaying "Good day" if sales exceed 100 meals, "Low day" otherwise. (b) Write the formula counting the number of good days. (c) Which chart type best shows the evolution of sales during the week? Justify.

SOLUTION

Solution.

(a) In C2: `=IF(B2>100;"Good day";"Low day")`, then copy down to C8. The test `B2>100` is evaluated for each row; because B2 is relative, it adapts to each day.

(b) `=COUNTIF(C2:C8;"Good day")` counts the cells of the range containing exactly the text "Good day". One could also count directly on the numbers: `=COUNTIF(B2:B8;">100")`.

(c) A line chart is the most appropriate: the days form an ordered time sequence (Mon → Sun) and a line emphasises the evolution (rises and falls) of sales during the week. A column chart would also be acceptable for comparing the days, but a pie chart would be wrong here, because a pie chart shows shares of a total, not an evolution.

Key point: choose the chart according to the message: comparison → columns, proportion → pie, evolution → line.

8.5 Linking Worksheets

METHOD **Linking worksheets and workbooks**

1. Same workbook, different sheet: `=SheetName!CellReference`. If the sheet name contains spaces or special characters, enclose in single quotes: `='Term 1'!B5`.

2. 3D reference (consolidation across adjacent sheets with identical layout): `=SUM(Sheet1:Sheet3!B2)` adds cell B2 from Sheet1, Sheet2, and Sheet3. Works for `SUM`, `AVERAGE`, `MAX`, `MIN`, `COUNT`.
3. Different workbook (external link): `=[WorkbookName.xlsx]SheetName!CellReference`. Example: `=[Budget2025.xlsx]Sheet1!B5`. The source workbook must be open or the path included.
4. Updating links: when the source changes, the linked cell recalculates. On opening a workbook with external links, Excel asks "Update links?" → click `Update`.
5. Break links (if needed): Data → Edit Links → Break Link → converts formulas to static values.

Reflex: linking avoids duplicating data — one single source, updated once, reflected everywhere. This is the principle of **data integrity**.

KEY POINTS **Linking syntax summary**

Situation	Syntax example
Same sheet	`=B2`
Different sheet, no spaces	`=Term1!B2`
Different sheet, with spaces	`='Term 1'!B2`
3D sum across Term1, Term2, Term3	`=SUM(Term1:Term3!B2)`
External workbook (open)	`=[Budget.xlsx]Sheet1!B2`
External workbook (closed, full path)	`='C:\Data\[Budget.xlsx]Sheet1'!B2`

WORKED EXAMPLE 11 **Consolidating term results — GCE style**

A school keeps three worksheets in one workbook: `Term1`, `Term2`, `Term3`, each with the same layout; the total marks of each student are in column F (F2 for the first student). A summary sheet `Annual` must show, for the first student, the annual total and the annual average. Write the required formulas.

SOLUTION

Solution.

1. Annual total in `Annual!B2`: `=Term1!F2+Term2!F2+Term3!F2`, or more compactly with a 3D sum: `=SUM(Term1:Term3!F2)`.

2. Annual average in `Annual!C2`: `=B2/3` or directly `=AVERAGE(Term1:Term3!F2)`.
3. Copy the formulas down for all students: the row number adapts automatically (relative references), while the sheet names remain unchanged.

Key point: the syntax `Sheet!Cell` is the heart of linking; the 3D form `Sheet1:Sheet3!Cell` is powerful when all sheets share the same structure.

8.6 What-If Analysis

EXPERIMENT **The "reverse" problem**

A student knows her CA mark (14/20) and the formula `Final = 0.4*CA + 0.6*Exam`. She wants a final mark of 12. What exam mark does she need? She could guess-and-check, or solve the equation. **Goal Seek** does the algebra for her. This is **what-if analysis**: changing inputs to hit a target output.

METHOD **Performing a what-if analysis**

What-if analysis consists of changing input values to observe the effect on results computed by formulas.

1. Build the model with formulas (never typed results), so that outputs depend on inputs. Identify: input cells (variables you can change), formula cells (calculated results).
2. Manual what-if: change an input cell and observe the recalculated outputs. Good for quick exploration.
3. Goal Seek (Data → What-If Analysis → Goal Seek): finds the input value needed to reach a target result. Fill in:
   - Set cell: the formula cell (the result you want to hit).
   - To value: the target number.
   - By changing cell: the input cell you can vary.
4. Data Table (one or two variables): shows the result of a formula for a whole series of input values. Data → What-If Analysis → Data Table.
5. Scenario Manager: save and compare several sets of input values (e.g. Optimistic / Normal / Pessimistic budgets). Data → What-If Analysis → Scenario Manager.

Reflex: Goal Seek requires that the "Set cell" contains a formula depending (directly or indirectly) on the "By changing cell".

KEY POINTS **What-if tools comparison**

Tool	Use when…	Output
Manual change	Quick test, one or two values	Immediate recalculation
Goal Seek	One target, one unknown input	Single solution value
Data Table (1 var)	See results for many values of one input	Column/row of results
Data Table (2 var)	See results for combinations of two inputs	Matrix of results
Scenario Manager	Compare a few named scenarios (Best/Worst)	Switchable named sets

WARNING **Goal Seek traps**

- "Set cell" must be a formula, not a constant.
- "By changing cell" must be an input (no formula), and must affect the set cell.
- Goal Seek finds one solution; non-linear models may have multiple solutions.
- It uses numerical iteration; result may be approximate (e.g. 10.6666667). Round appropriately for context.

WORKED EXAMPLE 12 **Goal Seek: mark needed at the GCE**

A candidate's final mark is computed as `Final = 0.4*CA + 0.6*Exam`, where CA (continuous assessment, in B1) is already known: 14/20, and the Exam mark is in B2. The final mark is in B3. (a) Write the formula in B3. (b) The candidate wants a final mark of at least 12/20. Use Goal Seek to find the minimum exam mark. (c) Compute the result by hand to verify.

SOLUTION

Solution.

(a) In B3: `=0.4*B1+0.6*B2`.

(b) Data → What-If Analysis → Goal Seek: Set cell = `B3`; To value = `12`; By changing cell = `B2`. Click `OK`. Goal Seek reports the value B2 ≈ 10.6667. The candidate therefore needs at least about 10.67/20 at the exam (round up to 11 if marks are integers).

(c) Verification by algebra:

$$0.4 \times 14 + 0.6 \times x = 12$$
$$5.6 + 0.6x = 12$$
$$0.6x = 6.4$$
$$x = \frac{6.4}{0.6} = \frac{64}{6} = \frac{32}{3} \approx 10.67$$

The spreadsheet result is confirmed.

Key point: Goal Seek automates exactly this "reverse" calculation; it only works because B3 is a formula linked to B2.

8.7 Integration Activity: from Data to a Complete Report

EXPERIMENT **The real GCE Paper 3 task**

The practical exam gives you: a raw data file (CSV or Excel), a word processor template, and a list of requirements: "Compute X, create chart Y, write a report with title, TOC, conclusions, page numbers, export to PDF." You have 2h30. The only way to finish is a rehearsed workflow: Spreadsheet first (clean data, formulas, chart), then Word (structure, paste linked objects, finalise).

METHOD **Integrating spreadsheet and word processor — full workflow**

Typical GCE integration task: analyse data in a spreadsheet, then present conclusions in a word-processed document.

1. Spreadsheet part:
   (a) Import/open raw data; convert to Table (`Ctrl+T`); clean (remove duplicates, fix text-in-numbers).
   (b) Compute required columns with formulas (use $ for constants).
   (c) Apply logical functions (`IF`, `COUNTIF`, `SUMIF`, `VLOOKUP`) as asked.
   (d) Create the required chart: select correct range → correct chart type → add titles, labels.
   (e) Verify: test formulas with simple values; check copy behaviour (relative/absolute).
2. Transfer to word processor:
   (a) In spreadsheet: select table → `Ctrl+C`.

(b) In word processor: Home → Paste → Paste Special → Paste Link → Microsoft Excel Worksheet Object (or `LibreOffice Spreadsheet`). This keeps the document synchronised.

(c) For chart: click chart → `Ctrl+C` → in Word: Paste Special → Paste Link → Chart Object.

3. Document part:

(a) Apply Title, Heading 1, Heading 2, Normal styles.

(b) Write introduction, methodology, findings (referencing the linked table/chart), conclusion.

(c) Insert automatic Table of Contents (References → Table of Contents).

(d) Insert page numbers, headers/footers (name, candidate number, date).

(e) `Ctrl+A` → `F9` to update all fields.

(f) Save `.docx`, then Export to PDF.

Reflex: a linked paste keeps the document synchronised with the spreadsheet; a simple paste freezes a copy. In the exam, always use Paste Link unless explicitly told "paste as picture".

KEY POINTS **Integration checklist — tick before submitting**

- ☐ Spreadsheet: all formulas use cell references (no typed numbers), $ used correctly.
- ☐ Spreadsheet: chart type matches the question (evolution → line, comparison → column, proportion → pie).
- ☐ Word: heading styles used → TOC generates correctly.
- ☐ Word: table and chart pasted as links (right-click → Linked Worksheet Object → Update Link works).
- ☐ Word: page numbers, footer with name/candidate number, title page.
- ☐ Both files saved (`.xlsx`, `.docx`) and PDF exported.

WORKED EXAMPLE 13 **Complete integration task: mobile money agency**

An MTN Mobile Money agency in Douala records, in a worksheet, for each of 6 agents: the number of daily transactions (column B) and the total amount in FCFA (column C). You

must: (1) compute in D2 the average amount per transaction for the first agent; (2) display in E2 "Bonus" if the agent made more than 50 transactions, "No bonus" otherwise; (3) count the agents receiving a bonus; (4) produce a column chart comparing the agents' transaction numbers; (5) insert the table and the chart into a one-page word-processed report with a title, a short conclusion, and your name in the footer.

SOLUTION

Solution.

1. In D2: `=C2/B2` (amount divided by number of transactions), formatted as Currency (FCFA, 0 decimals) via Format Cells → Currency, then copied down to D7.
2. In E2: `=IF(B2>50;"Bonus";"No bonus")`, copied down to E7.
3. In a cell below, e.g. E9: `=COUNTIF(E2:E7;"Bonus")`.
4. Select the agents' names (A1:A7) and the transaction column (B1:B7) → Insert → Chart → Clustered Column. Add Chart Title: "Daily Transactions per Agent", Horizontal Axis Title: "Agent", Vertical Axis Title: "Number of Transactions". Remove legend (single series).
5. In the word processor:
   (a) Type the title "MTN Mobile Money Douala — Daily Performance Report" with Title style.
   (b) Write one short paragraph of conclusion, e.g.: "Agent 3 (Ngono) recorded the highest transactions (62) and qualifies for bonus. Three out of six agents exceeded the 50-transaction threshold. The agency should investigate Agent 2's low volume (28)."
   (c) Copy the table (A1:E7) from Excel → in Word: Paste Special → Paste Link → Microsoft Excel Worksheet Object. Resize to fit page width.
   (d) Copy the chart from Excel → in Word: Paste Special → Paste Link → Chart Object. Position below the table.
   (e) Insert footer: Insert → Footer → Blank → type your name and candidate number.
   (f) Save as `MM_Douala_Report.docx`, update fields (`F9`), export to PDF `MM_Douala_Report.pdf`.

Key point: this task chains all the skills of the chapter — formulae, logical functions, conditional counting, charts, linking and document production — exactly as in the integration activities of the syllabus (Lesson 64).

EXERCISE 14 **Practice: End-to-end GCE simulation**

You are given a file `ExamData.xlsx` with two sheets: `RawMarks` (columns: RegNo, Name, Class, Paper1, Paper2, Paper3) and `Coeffs` (Paper1=2, Paper2=3, Paper3=5 in B2:B4). In a new sheet `Results`:

1. Link the student data from `RawMarks` (use `=RawMarks!A2` etc.).
2. Compute `WeightedTotal` = Paper1*2 + Paper2*3 + Paper3*5 using absolute references to `Coeffs!B2:B4`.
3. Compute `Average` = WeightedTotal / 10 (sum of coefficients).
4. In `Decision` column: `=IF(Average>=12;"Pass";IF(Average>=10;"Probation";"Fail"))`.
5. Count Pass, Probation, Fail with `COUNTIF`.
6. Create a column chart of Average per student (sorted descending).
7. Copy the Results table and chart into a Word report with title, TOC, conclusion, page numbers, footer with your name. Export to PDF.

Time yourself: target 45 minutes for spreadsheet, 30 minutes for report.

SOLUTION **Exercise: End-to-end GCE simulation**

1. In `Results!A2`: `=RawMarks!A2`, copy across to D2 (Name, Class), then down.
2. In `Results!F2` (WeightedTotal): `=C2*$Coeffs.$B$2+D2*$Coeffs.$B$3+E2*$Coeffs.$B$4` (assuming Paper1 in C, Paper2 in D, Paper3 in E). Copy down.
3. In `Results!G2` (Average): `=F2/SUM($Coeffs.$B$2:$Coeffs.$B$4)`. Copy down.
4. In `Results!H2` (Decision): `=IF(G2>=12;"Pass";IF(G2>=10;"Probation";"Fail"))`. Copy down.
5. Counts: `=COUNTIF(H2:H100;"Pass")`, `=COUNTIF(H2:H100;"Probation")`, `=COUNTIF(H2:H100;"Fail")`.
6. Chart: sort `Results` by Average descending (Data → Sort), select Name and Average columns → Insert → Column Chart.
7. Word: follow Method 7 workflow. Verify linked objects update when you change a mark in `RawMarks`.

9 EXERCISES

9.1 I check my knowledge

EXERCISE 15 MCQ on cell referencing

Which of the following formulas, when copied from cell E2 to F4, will keep the reference to cell C1 fixed?

1. =B2*C1+D2
2. =B2*$C$1+$D2
3. =$B$2*C1+D$2
4. =B2*C$1+$D$2

EXERCISE 16 True/False with justification

State whether each statement is true or false and justify your answer.

1. In a spreadsheet, a relative reference changes when the formula is copied to another cell.
2. The function SUMIF can only be used with numeric criteria.
3. A chart of type “Pie” is the most appropriate to show the evolution of monthly income over a year.
4. The error #NAME? appears when a formula tries to divide by zero.

EXERCISE 17 Definitions

Define the following terms as used in spreadsheet applications:

1. Absolute cell reference
2. Mixed cell reference
3. What-If Analysis
4. Mail merge

EXERCISE 18 **Direct calculations**

A student obtained the following marks in three sequences: 8 (weight 1), 9 (weight 1), 11 (weight 2). Compute the weighted average.

9.2 I practise

EXERCISE 19 **SUMIF formula for fee sheet**

A worksheet contains 50 rows with columns: A (Name), B (Class), C (Amount Paid). Write the formula that calculates the total amount paid by students in "Form Five" only.

EXERCISE 20 **Relative vs absolute referencing**

Explain the difference between relative and absolute referencing. Give one practical situation where each type is required.

EXERCISE 21 **Goal Seek scenario**

A student has averages of 8, 9 and 11 in three sequences weighted 1, 1 and 2 respectively. Using the logic of Goal Seek, determine the mark needed in a fourth sequence (weight 2) to achieve an overall weighted average of 10. Show the equation you would set up.

EXERCISE 22 **Chart selection**

For each of the following data sets, choose the most appropriate chart type and justify your choice:

1. Monthly school income over a 12-month period.
2. Percentage share of students per class (Form 1 to Upper Sixth).
3. Comparison of boys' and girls' average marks in five subjects.

9.3 I prepare for the GCE

EXERCISE 23 **Mail merge procedure (10 marks)**

List, in the correct order, the steps required to produce 200 personalised fee-reminder letters using mail merge from an Excel data source. Name the three main components involved in the mail merge process.

EXERCISE 24 **Error diagnosis (12 marks)**

A worksheet displays the error #DIV/0! in the "Average" column and #NAME? in another cell.

1. Diagnose the probable cause of each error.
2. Propose a correction for each error.

EXERCISE 25 **Linked worksheets and renaming (13 marks)**

A school keeps term results on three separate sheets named "Term1", "Term2" and "Term3". On a "Summary" sheet you want to add a student's three term totals.

1. Write the formula on the "Summary" sheet that adds the three term totals for a student whose total is in cell B10 on each term sheet.
2. Explain what happens to the link if the "Term2" sheet is renamed to "SecondTerm".
3. Suggest a method to avoid broken links when sheets are renamed.

10 SOLUTIONS TO THE EXERCISES

10.1 Solutions – I check my knowledge

SOLUTION **Exercise 1 – MCQ on cell referencing**

Correct answer: (b) =B2*$C$1+$D2
Justification. The requirement is that the reference to C1 must remain fixed when the formula is copied from E2 to F4. A reference is fixed only if both its column letter and its row number are preceded by the dollar sign $, i.e. $C$1 (**absolute reference**).

- Option (a): C1 is fully relative; copied from E2 to F4 (one column right, two rows down) it becomes D3. Wrong.
- Option (b): C1 is written $C$1, so it stays C1 wherever the formula is copied. Correct.
- Option (c): C1 is fully relative (it becomes D3); only D$2 is partly fixed. Wrong.
- Option (d): C$1 fixes only the row (1), but the column C changes to D when copied one column to the right, giving D$1. Wrong.

Note that in option (b) the term $D2 is a **mixed reference**: the column D is fixed, the row is free. This does not affect the question, which concerns only C1.

SOLUTION **Exercise 2 – True/False with justification**

1. TRUE. By definition, a **relative reference** (e.g. B2) is interpreted as a position relative to the cell containing the formula. When the formula is copied, the reference is automatically adjusted by the same displacement: copying =B2 from E2 down to E3 gives =B3.
2. FALSE. **SUMIF** also accepts text criteria and criteria with comparison operators, e.g. =SUMIF(B2:B50;“Form Five”;C2:C50) or =SUMIF(C2:C50;“>50000”). The criterion is not restricted to numbers.
3. FALSE. A **pie chart** shows proportions of a whole at one moment; it cannot display evolution over time. To show the evolution of monthly income over a year, a **line chart** (or a column chart) is appropriate, with months on the horizontal axis.
4. FALSE. #NAME? appears when the spreadsheet does not recognise a name in the formula (misspelt function such as =SOMME(A1:A5) in an English version, or an undefined named range). Division by zero produces the error #DIV/0!.

SOLUTION **Exercise 3 – Definitions**

1. Absolute cell reference: a cell reference in which both the column and the row are locked with the dollar sign (e.g. $C$1). It always designates the same physical cell, whatever the destination of a copy of the formula.
2. Mixed cell reference: a reference in which only one of the two coordinates is locked: either the column ($C1) or the row (C$1). On copying, the locked part stays fixed while the free part is adjusted.
3. What-If Analysis: the set of spreadsheet techniques used to study how a change in input values affects the results of formulas. It includes tools such as Goal Seek (find the input needed to reach a target result), Scenario Manager and Data Tables.
4. Mail merge: a word-processing feature that automatically combines a main document (containing the fixed text and merge fields) with a data source (a table of recipients, e.g. an Excel sheet) to produce many personalised documents (letters, certificates, fee reminders), one per record.

SOLUTION **Exercise 4 – Direct calculations**

The weighted average is the sum of (mark × weight) divided by the sum of the weights:

$$\bar{x} = \frac{\sum x_i n_i}{\sum n_i} = \frac{8 \times 1 + 9 \times 1 + 11 \times 2}{1 + 1 + 2} = \frac{8 + 9 + 22}{4} = \frac{39}{4} = 9.75.$$

The weighted average is 9.75 out of 20.
In a spreadsheet, with marks in B2:B4 and weights in C2:C4, the same result is obtained by:

= SUMPRODUCT(B2:B4;C2:C4)/SUM(C2:C4)

Note: In English-language versions of Excel the list separator is a comma (,); in French versions or LibreOffice Calc it is a semicolon (;). Use the separator corresponding to your regional settings.

10.2 Solutions – I practise

SOLUTION **Exercise 5 – SUMIF formula for fee sheet**

The SUMIF function has the syntax `=SUMIF(range; criteria; sum_range)`. Here the class is tested in column B and the amounts to add are in column C:

= SUMIF(B2:B51;“Form Five”;C2:C51)

(50 data rows, from row 2 to row 51, assuming row 1 contains the headings.)
Remark: the text criterion must be written exactly as it appears in the sheet ("Form Five"), between double quotes. If the class name were stored in a cell, say E1, one would write `=SUMIF(B2:B51;E1;C2:C51)`, which is more flexible and avoids hard-coding.

SOLUTION **Exercise 6 – Relative vs absolute referencing**

Relative referencing (e.g. B2): the reference designates a cell by its position relative to the formula. When the formula is copied, the reference moves by the same displacement (copy one row down ⇒ B2 becomes B3).
Absolute referencing (e.g. $B$2): the reference designates a fixed cell; it never changes on copying.
Practical situations:

- Relative: computing each student's term total with `=SUM(B2:D2)` in E2, then copying the formula down for the whole class – each row must sum its own marks.
- Absolute: converting amounts in CFA francs using an exchange rate stored in a single cell H1: the formula `=B2*$H$1` copied down the column must always read the same rate cell. Similarly, applying a single school-fees amount or a VAT rate stored once to many rows.

SOLUTION **Exercise 7 – Goal Seek scenario**

Let x be the mark needed in the fourth sequence (weight 2). The overall weighted average must equal 10:

$$\frac{8\times 1+9\times 1+11\times 2+x\times 2}{1+1+2+2}=10$$

$$\frac{39+2x}{6}=10$$

$$39+2x=60$$

$$2x=21$$

$$x=10.5.$$

The student needs 10.5 out of 20 in the fourth sequence.
With Goal Seek in the spreadsheet: put the marks and weights in cells, compute the average in a cell (say F5) with `=SUMPRODUCT(...)/SUM(...)`, leaving the fourth mark cell empty. Then open Data → What-If Analysis → Goal Seek and set: Set cell = F5, To value = 10, By changing cell = the fourth mark cell. Goal Seek returns 10.5 automatically.

SOLUTION **Exercise 8 – Chart selection**

1. Monthly school income over 12 months: line chart (a column chart is also acceptable). Justification: a line chart is the standard tool for showing the evolution/trend of a quantity over time; the months go on the horizontal axis.
2. Percentage share of students per class: pie chart (or 100% stacked bar). Justification: the data are parts of a whole (all shares sum to 100%), which is exactly what a pie chart displays.
3. Boys' vs girls' average marks in five subjects: clustered (grouped) column/bar chart with two series. Justification: it allows a direct comparison of two categories across several items (subjects), with one column per sex side by side for each subject.

10.3 Solutions – I prepare for the GCE

SOLUTION **Exercise 9 – Mail merge procedure (10 marks)**

The three main components of a mail merge (3 marks):

1. The main document – the letter template containing the fixed text and the merge fields;
2. The data source – the Excel sheet (or CSV, Access table) with one row per student (Name, Class, Amount owed, …);
3. The merged document(s) – the 200 personalised letters produced by the combination.

Steps in the correct order (7 marks – 1 mark per correctly ordered step, any 7 of the following):

1. Prepare and save the Excel data source: one header row with field names (e.g. `Name`, `Class`, `Amount`), one record per student, no blank rows.
2. Open the word processor (MS Word or LibreOffice Writer) and create (or open) the main document; type the fixed text of the fee-reminder letter.
3. Start the mail merge: Mailings → Start Mail Merge → Letters (Word) or Tools → Mail Merge Wizard (LibreOffice).
4. Select the recipients: Select Recipients → Use an Existing List, then browse to the Excel file and choose the correct worksheet.
5. Insert the merge fields (e.g. `<<Name>>`, `<<Class>>`, `<<Amount>>`) at the appropriate positions in the letter.

6. (Optional) Set rules or filter/sort the recipient list if only some students should receive the letter (e.g. only those with Amount > 0).
7. Preview the results to check that the fields are correctly replaced for several records.
8. Complete the merge: Finish & Merge → Edit Individual Documents (produces all 200 letters in a new file) or Print Documents.
9. Save and/or print the 200 personalised letters.

Examiner expectations: the candidate must show that the data source is prepared before the connection, that merge fields are inserted before the merge, and must name the three components precisely. Vague answers such as "click mail merge" without the logical order lose marks.

SOLUTION **Exercise 10 – Error diagnosis (12 marks)**

(a) Diagnosis (6 marks – 3 per error):
#DIV/0! in the "Average" column: the formula attempts a division by zero or by an empty cell. Typical cause: `=SUM(B2:D2)/E2` where the divisor E2 (number of subjects or total coefficient) is empty or equal to 0 – for example a newly added student whose coefficient has not yet been entered, or a copied formula whose divisor reference points to a blank cell.
#NAME?: the spreadsheet does not recognise a text in the formula. Typical causes: a misspelt function name (`=AVEARGE(B2:D2)` or the French `=MOYENNE(...)` in an English version), a named range that does not exist or was deleted, or text entered in a formula without double quotes.
(b) Corrections (6 marks – 3 per error):
For #DIV/0!:

- Enter the missing value in the divisor cell, or correct the divisor reference; and/or
- Make the formula robust with a test: `=IF(E2=0;``'';SUM(B2:D2)/E2)` or `=IFERROR(SUM(B2:D2)/E2;``--'')`, so that the sheet stays clean while data are incomplete.

For #NAME?:

- Check and correct the spelling of the function (use the function wizard or autocomplete);
- Recreate or correct the named range (Formulas → Name Manager);
- Put any literal text used in the formula between double quotes.

Examiner expectations: one mark for identifying the error type, one for a plausible concrete cause, one for a valid correction, per error. Simply restating "division by zero" for #DIV/0! without proposing a correction earns only partial credit.

SOLUTION **Exercise 11 – Linked worksheets and renaming (13 marks)**

(a) Formula on the Summary sheet (4 marks):
A reference to another sheet uses the syntax SheetName!Cell. The formula that adds the three term totals is:

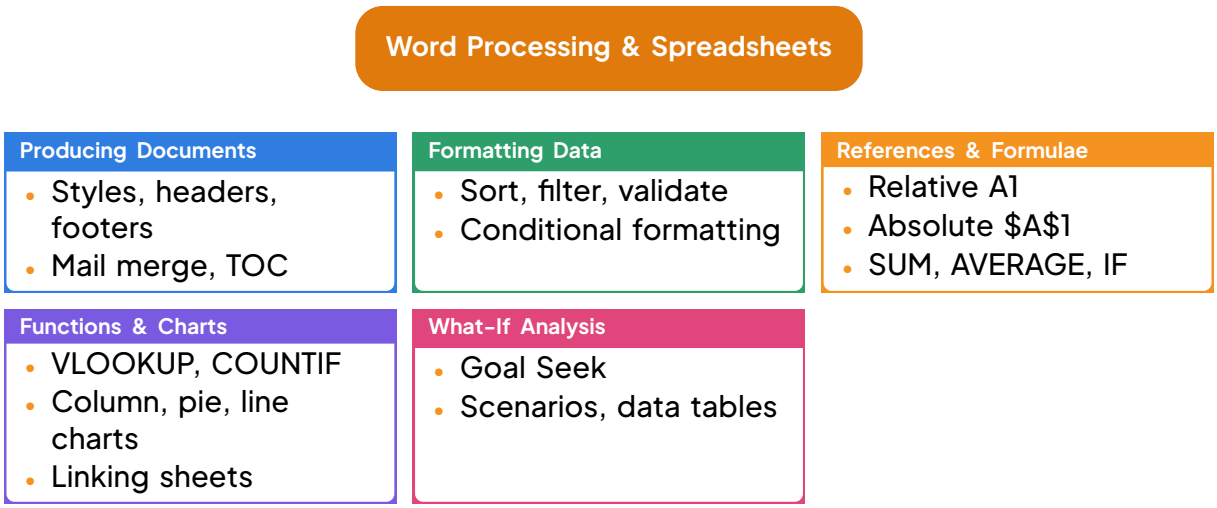
= Term1!B10 + Term2!B10 + Term3!B10

Equivalently, using a 3-D reference across consecutive sheets: `=SUM(Term1:Term3!B10)`.
(2 marks for correct sheet!cell syntax, 2 marks for adding the three correct cells.)
(b) Effect of renaming "Term2" to "SecondTerm" (4 marks):
In modern spreadsheets (Excel, LibreOffice Calc), renaming a sheet does not break the link: the application automatically updates all formulas that reference the sheet, so the formula becomes `=Term1!B10+SecondTerm!B10+Term3!B10` and the result is unchanged. However, if the sheet is deleted (or the reference is typed as plain text in an external file path), the link breaks and the formula returns the error #REF!. (2 marks for stating the automatic update, 2 marks for mentioning the #REF! risk in the deletion case.)
(c) Method to avoid broken links (5 marks – any well-developed method, e.g.):

- Use named ranges (e.g. define the name TotalTerm2 for `Term2!$B$10`) and write formulas with the names: `=TotalTerm1+TotalTerm2+TotalTerm3`. Names refer to cells, not to sheet labels typed in formulas, so maintenance is easier and formulas are more readable.
- Finalise sheet names before building the summary formulas, and never delete a source sheet – hide it instead if necessary.
- Keep all linked sheets in the same workbook rather than linking to external files, and avoid moving/renaming the file or its folder.

Examiner expectations: precise syntax with the exclamation mark in (a); in (b) the candidate must distinguish renaming (link updated automatically) from deleting (#REF!); in (c) any coherent, technically valid preventive method with a brief explanation earns full marks.

11 SUMMARY SHEET



Mind map of the chapter: from document production to what-if analysis.

FICHE BILAN **L'essentiel à connaître pour l'examen**

Key definitions

- A **word processor** (MS Word, LibreOffice Writer) is software used to create, edit, format, save and print text documents.
- **Formatting** means changing the appearance of a document: character formatting (font, size, bold, colour), paragraph formatting (alignment, line spacing, indentation) and page formatting (margins, orientation, headers and footers).
- A **style** is a named set of formatting attributes applied in one click; modifying the style updates the whole document automatically.
- **Mail merge** combines a main document (letter) with a data source (list of names) to produce personalised documents.
- A **spreadsheet** (MS Excel, LibreOffice Calc) is a grid of **cells** organised in rows (numbered) and columns (lettered); the intersection $B3$ is a cell, a rectangular block like $A1:D10$ is a **range**.
- A **formula** always begins with = and combines values, cell references and functions; a **function** is a predefined formula such as `SUM` or `AVERAGE`.
- **What-if analysis** is the technique of changing input values to observe the effect on results computed by formulae.

Formulae and functions to know

- Basic: `=SUM(A1:A10)`, `=AVERAGE(B2:B20)`, `=MAX(...)`, `=MIN(...)`, `=COUNT(...)` (numbers only), `=COUNTA(...)` (non-empty cells).
- Logic: `=IF(C2>=10,"Admitted","Failed")` — syntax `IF(test; value_if_true; value_if_false)`.
- Advanced: `=COUNTIF(A2:A50,"M")`, `=SUMIF(B2:B50,">10000")`, `=VLOOKUP(code; table; column; FALSE)`.
- Referencing: relative `A1` changes when copied; absolute `$A$1` never changes; mixed `$A1` or `A$1` locks only the column or the row.
- Linking worksheets: `=Sheet2!B5` reads a cell from another sheet; `='[Book2.xlsx]Sheet1'!A1` reads from another workbook.
- Percentage of a total: share = $\frac{\text{part}}{\text{total}} \times 100$.

Methods

- Producing a document: type the text → save early → apply character and paragraph formatting → insert page numbers, header/footer → use heading styles → generate the table of contents → proofread and print preview.
- Building a marksheet: enter labels and data → write one formula with correct references → copy it down with the fill handle → format (borders, number format, conditional formatting) → sort or filter if required.
- Goal Seek: Data → What-If Analysis → Goal Seek: set the result cell, to the target value, by changing the input cell.
- Charts: select the data (with labels) → Insert → choose chart type (column for comparisons, pie for shares, line for trends) → add title and axis labels.

Common mistakes

- Forgetting the = sign: the entry is stored as text, not computed.
- Copying a formula with a relative reference where an absolute one was needed (e.g. a single VAT cell must be `$B$1`).
- Confusing `COUNT` (counts numbers) with `COUNTA` (counts all non-empty cells).
- Using a pie chart for time series, or a chart without title and labels.
- Typing results by hand instead of using formulae: the sheet no longer updates when data change.
- In mail merge, forgetting to link the data source before inserting merge fields.

GCE tips

- Read the question twice: the exam often asks for the exact formula to write in a named cell — quote it fully, e.g. `=IF(D2>=50,"Pass","Fail")`.

- State the chart type and justify the choice in one sentence.
- For practical papers, save your work regularly with the exact file name requested.
- Know the vocabulary: cell, range, worksheet, workbook, fill handle, sorting key, filtering criterion.
- When asked to distinguish, contrast in two clear columns (e.g. relative vs absolute referencing).

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